Module 1: Foundations of *k*-mer Based Genome Profiling

Kamil S. Jaron

October 2026



Scalable Genomics and Pangenomics 2026

Time/Date	11th October Sunday	12th October Monday	13th October Tuesday	14th October Wednesday	15th October Thursday	16th October Friday	Time
08:30 - 08:45							08:30 - 08:45
08:45 - 09:00							08:45 - 09:00
09:00 - 09:10		Participants Check In - Montmirador	Participants Check In - Montmirador	Participants Check In - Montmirador	Participants Check In - Montmirador	Introduce Group Projects (15 mins)	09:00 - 09:10
09:10 - 09:25		Lecture: Foundations of K-mer Based Genome Analysis (Kamil Jaron)	Hands-on: Manipulating K-mers (Kamil Jaron, Camila Gutierrez Gonzalez)	Lecture: Building Pangenomics (Katie Jenike, Miriam Shmukarovic)	Case study discussion (Katie Jenike, Miriam Shmukarovic)	Work on Group Projects	09:10 - 09:25
09:25 - 09:40							09:25 - 09:40
09:40 - 10:00							09:40 - 10:00
10:00 - 10:15		Break	Break	Break	Break	Break	10:00 - 10:15
10:15 - 10:30							10:15 - 10:30
10:30 - 10:45		Practical: Genome Profiling Practical Using GenomeScope (Kamil Jaron)	Practical: Manipulating k-mers in a graph using Fastk, Mergury and Genomescope (Kamil Jaron, Camila Gutierrez Gonzalez)	Practical: Pangenomes and intraspecific variation (Miriam Shmukarovic, Katie Jenike)	Lecture (Richard Durbin) Genome Comparison using FastGA/FasTan (Richard Durbin)	Work on Group Projects	10:30 - 10:45
10:45 - 11:00							10:45 - 11:00
11:00 - 11:15							11:00 - 11:15
11:15 - 11:30							11:15 - 11:30
11:30 - 11:45						Group Project Presentations	11:30 - 11:45
11:45 - 12:00							11:45 - 12:00
12:00 - 12:15	Registration and Lunch at Rinaton Hall Restaurant	Lunch (now/here)	Lunch (now/here)	Lunch (now/here)	Lunch (now/here)	Wrap Up and Close Course	12:00 - 12:15
12:15 - 12:30							12:15 - 12:30
12:30 - 12:45						Lunch (grab bags)	12:30 - 12:45
12:45 - 13:00							12:45 - 13:00
13:00 - 13:15	Welcome Intros						13:00 - 13:15
13:15 - 13:30							13:15 - 13:30
13:30 - 13:45	Participants Introduction	Practical: Basics of k-mers and De Bruijn graphs (Katie Jenike)	Lecture: K-mers in a graph (Antoine Limasset, Camila Gutierrez Gonzalez)	Practical: Practical pangenomics (Miriam Shmukarovic, Katie Jenike)	Practical: Competitive Genome (Miriam Shmukarovic)		13:30 - 13:45
13:45 - 14:00							13:45 - 14:00
14:00 - 14:15	Lecture: Introduction to K-mer (Katie Jenike)						14:00 - 14:15
14:15 - 14:30						Training Team Leading Session Critical	14:15 - 14:30
14:30 - 14:45				Break	Break		14:30 - 14:45
14:45 - 15:00						Course Participants	14:45 - 15:00
15:00 - 15:15	Break	Break	Break	Guest Lecture (Ben Langmead)		Scientific Training & Events Team	15:00 - 15:15
15:15 - 15:30						Kamil Jaron	15:15 - 15:30
15:30 - 15:45	Lecture: Introduction to Pangenomics (Katie Jenike, Miriam Shmukarovic)	Lecture: Counting and Manipulating k-mers using FastK (Kamil Jaron, using Gene Myers's slides)	Practical: K-mers in a graph (Antoine Limasset, Camila Gutierrez Gonzalez)	Practical: sketching (Miriam Shmukarovic, Katie Jenike)	Practical: Reference Genome Comparison (Katie Jenike)	Katie Jenike	15:30 - 15:45
15:45 - 16:00						Antoine Limasset	15:45 - 16:00
16:00 - 16:15						Miriam Shmukarovic	16:00 - 16:15
16:15 - 16:30						Guest Speakers	16:15 - 16:30
16:30 - 16:45							16:30 - 16:45
16:45 - 17:00							16:45 - 17:00
17:00 - 17:15	Speed Networking and Informatics Problem Solving						17:00 - 17:15
17:15 - 17:30							17:15 - 17:30
17:30 - 17:45							17:30 - 17:45
17:45 - 18:00							17:45 - 18:00
18:00 - 18:15		Dinner	Dinner	Dinner	Dinner	Dinner	18:00 - 18:15
18:15 - 18:30							18:15 - 18:30
18:30 - 18:45	Welcome dinner	12th October Monday	13th October Tuesday	14th October Wednesday	15th October Thursday	16th October Friday	18:30 - 18:45

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09:10 - 09:25		Lecture: Foundations of K-mer Based Genome Analysis (Kamil Jaron)	Demo: Manipulating K-mers (Kamil Jaron, Camila Gutierrez Gonzalez)	Lecture: Building Pangenomics (Katie Jenike, Miriam Shmakulenac)	Case study discussion (Katie Jenike, Miriam Shmakulenac)	Work on Group Projects	09:10 - 09:25
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10:00 - 10:15							10:00 - 10:15
10:15 - 10:30							10:15 - 10:30
10:30 - 10:45		Practical: Genome Profiling Practical Using GenomeScope (Kamil Jaron)	Practical: Manipulating k-mers in Python using Yacrd, Meqsyy and Genalignplot (Kamil Jaron, Camila Gutierrez Gonzalez)	Practical: Pangenomes and intraspecific variation (Miriam Shmakulenac, Katie Jenike)	Lecture (Richard Durbin) Genome Comparison using FastGA/FasTAN (Richard Durbin)	Work on Group Projects	10:30 - 10:45
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15:45 - 16:00						Miriam Shmakulenac	15:45 - 16:00
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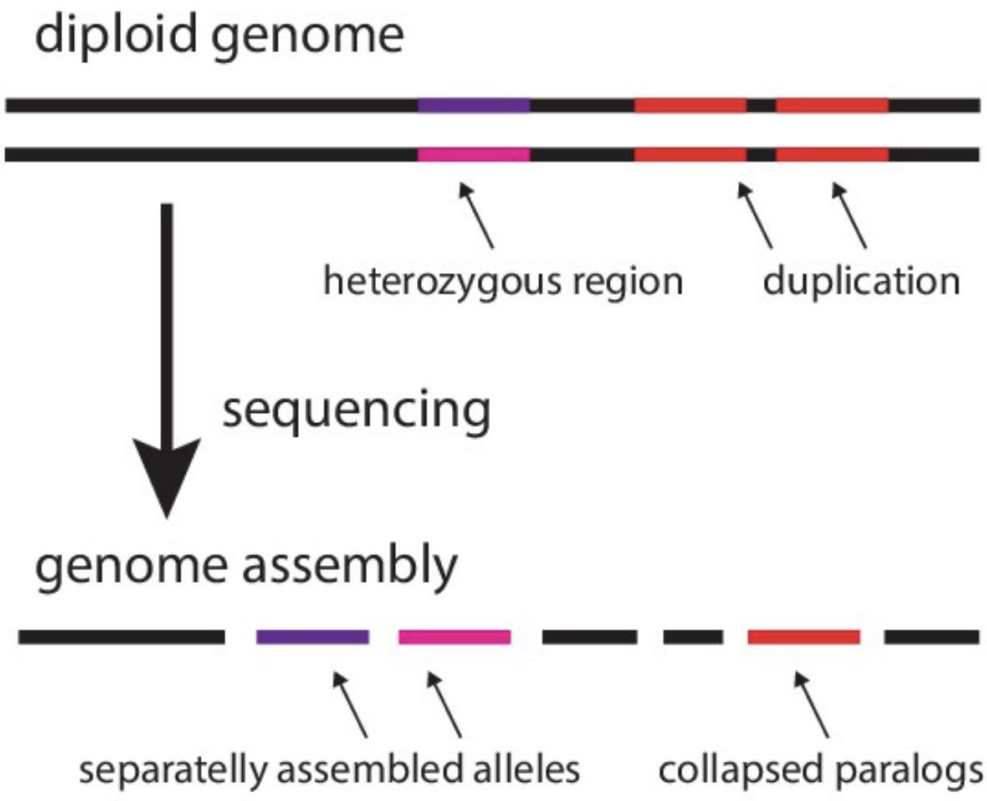
Genome sequencing



assembly



Assembly problems





Genome sequencing



assembly



genome profiling



13e6 individual plants in the harvest;
each weighted 843g and had 234 kernels on average

Part 1: Understanding *k*-mer histograms





for all reads in the bag…



ATCTAAACGATCGATCGATCGA read

ATCTAAA

TCTAAAC

CTAAACG

TAAACGA

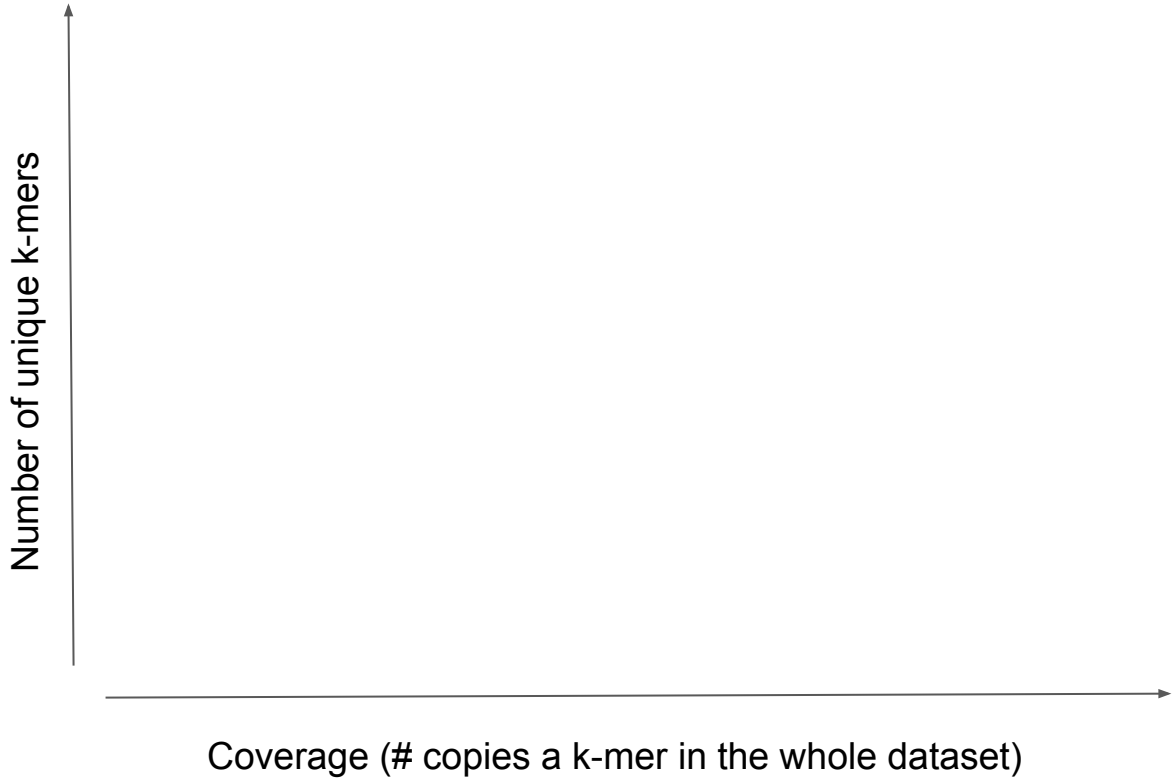
kmers (k = 7)

. . .

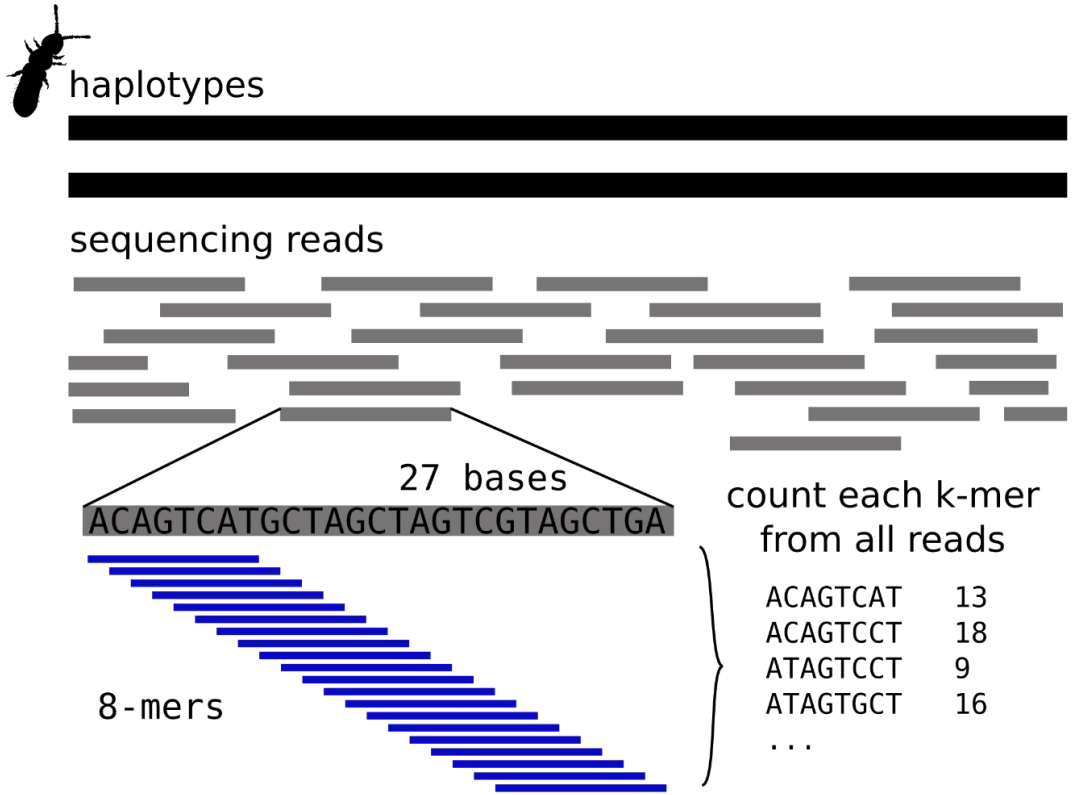
Counting frequency of all k-mers in all sequencing reads

kmer	coverage
AAAAAAAAAACAACGT	28
AAAAATAACACAACGT	31
AAAAATAACACAACGG	1
AAAAATAACGCAACGT	47
AAAATTTACGCAACGT	2
AAAATTTACGCAACGA	17
...	...

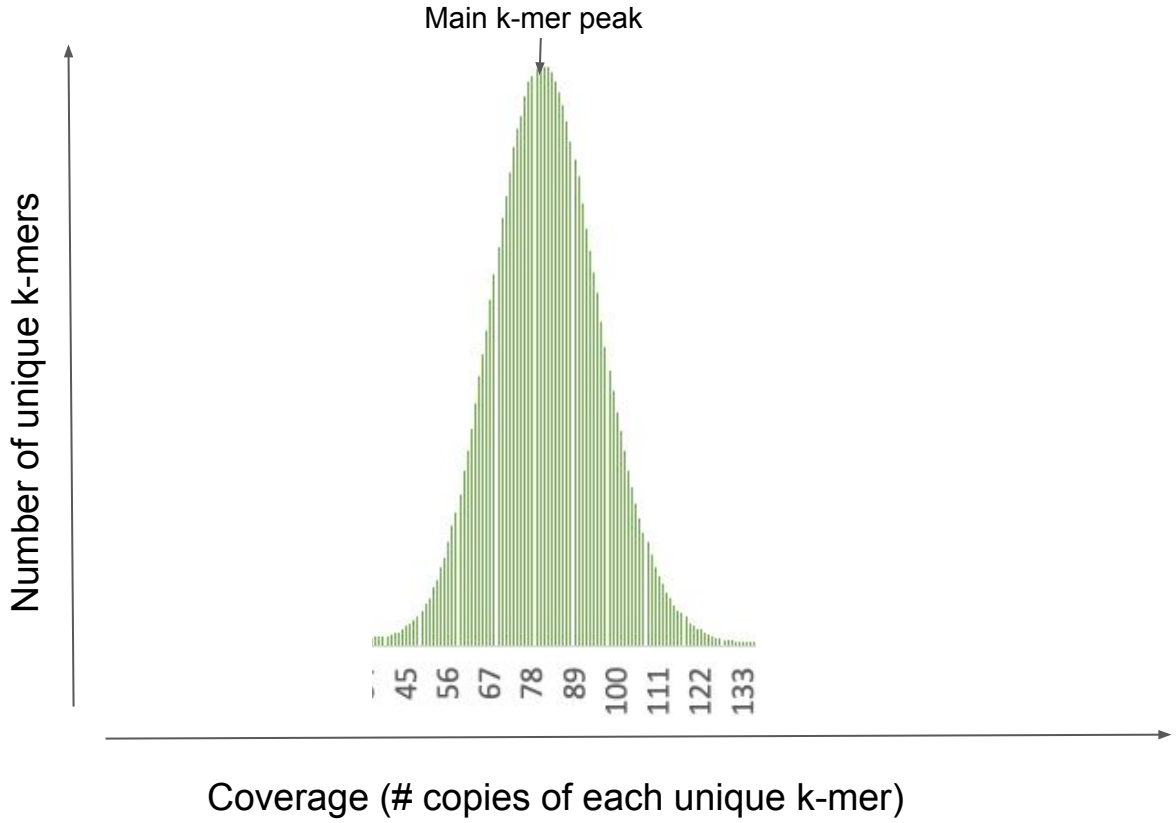
Plotting a k-mer histogram



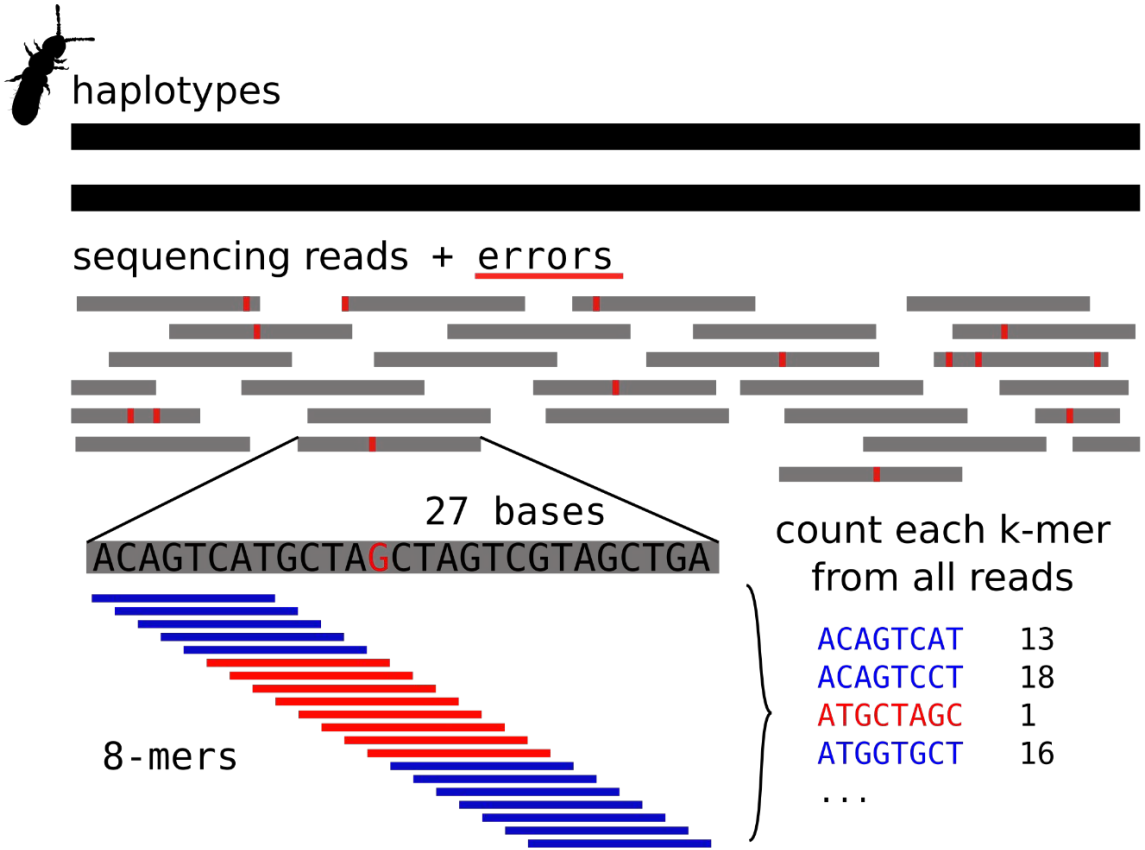
Plotting a perfect k-mer histogram



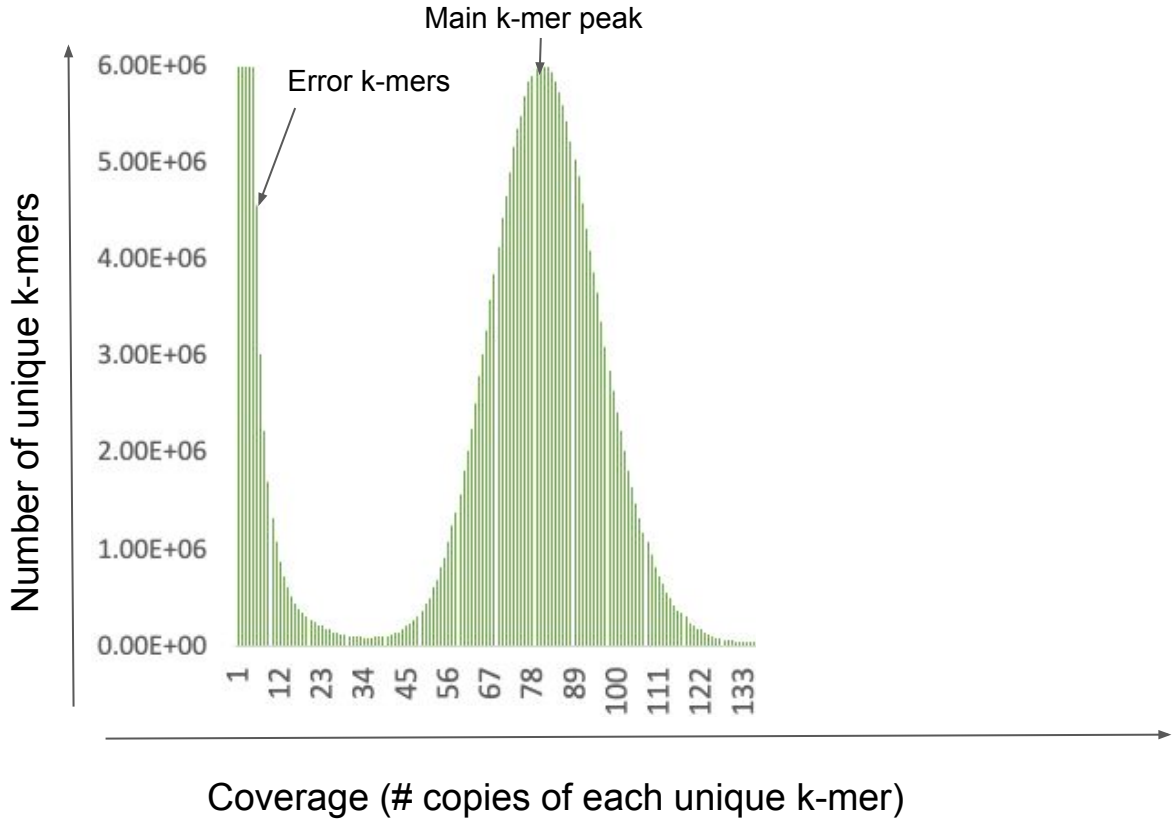
The perfect k-mer histogram



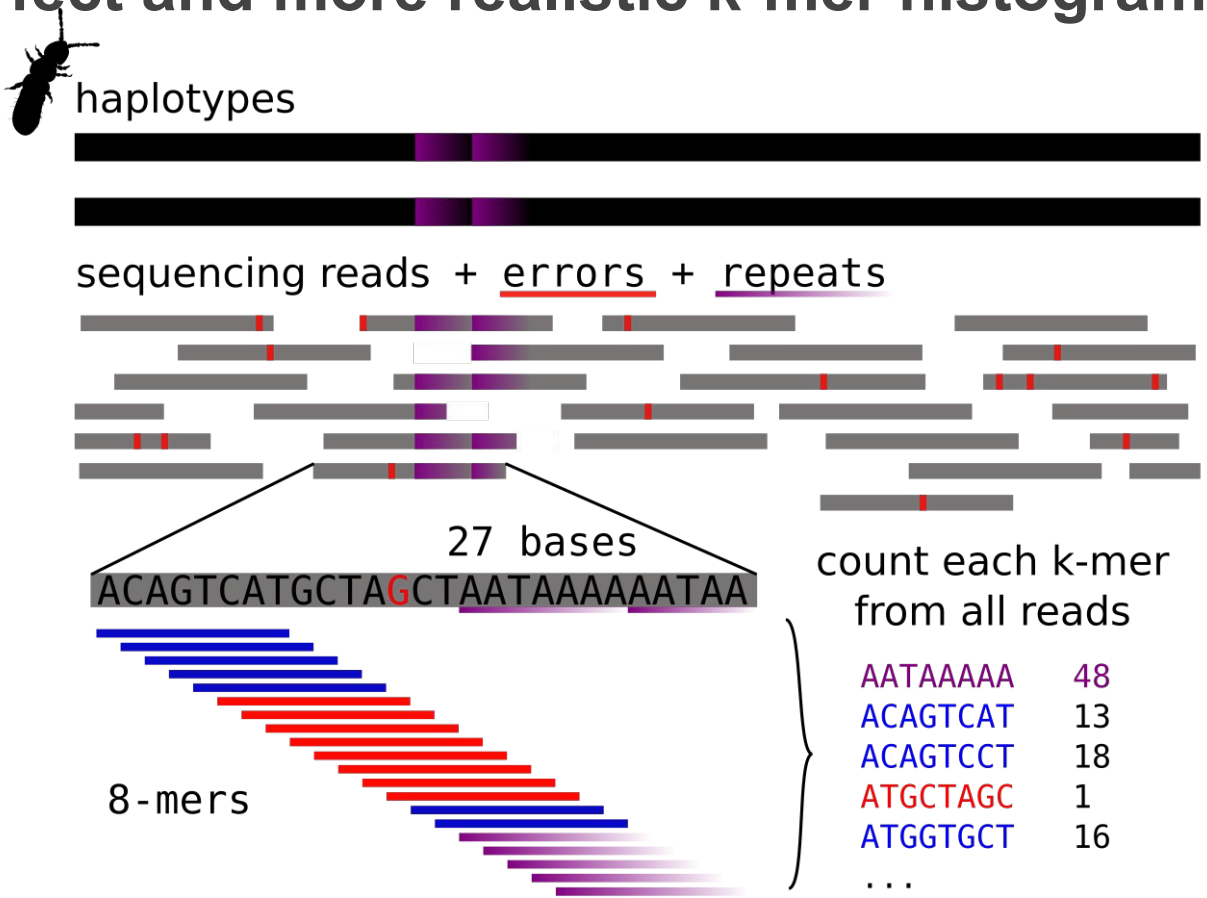
Creating an imperfect k-mer histogram



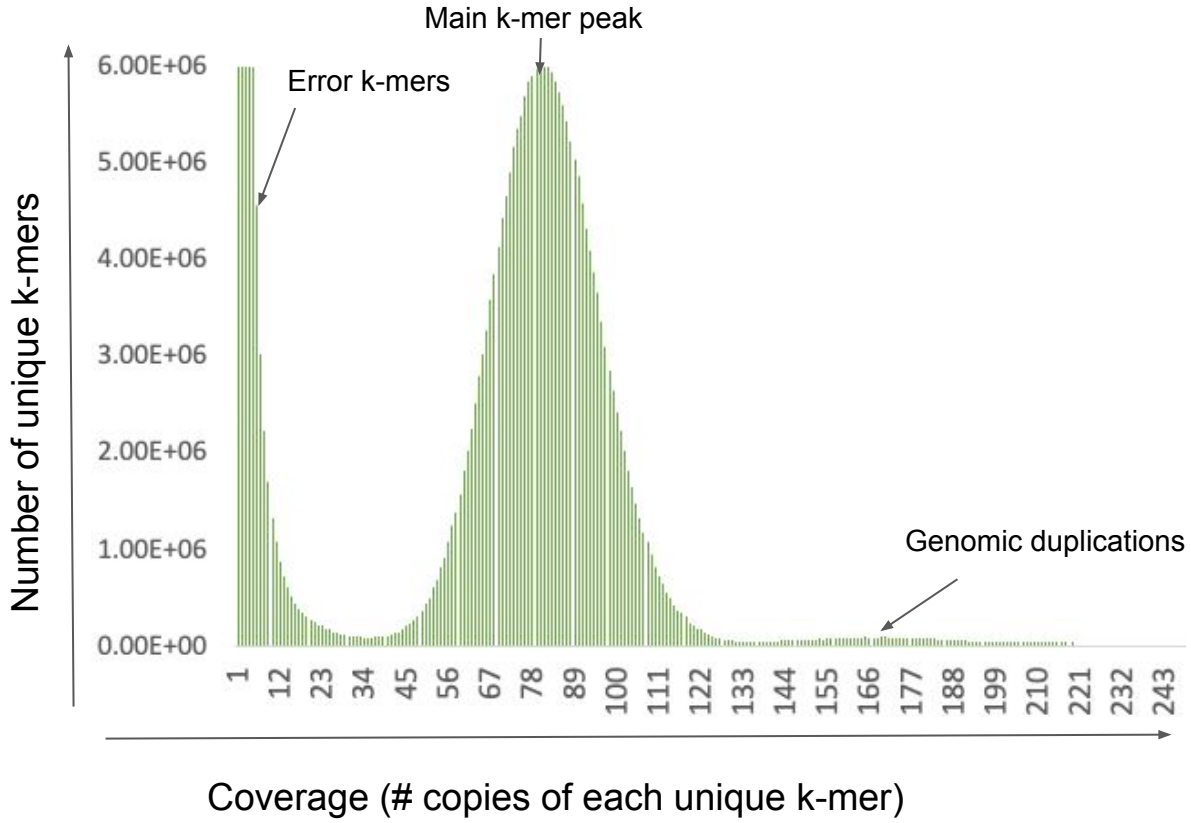
An imperfect histogram



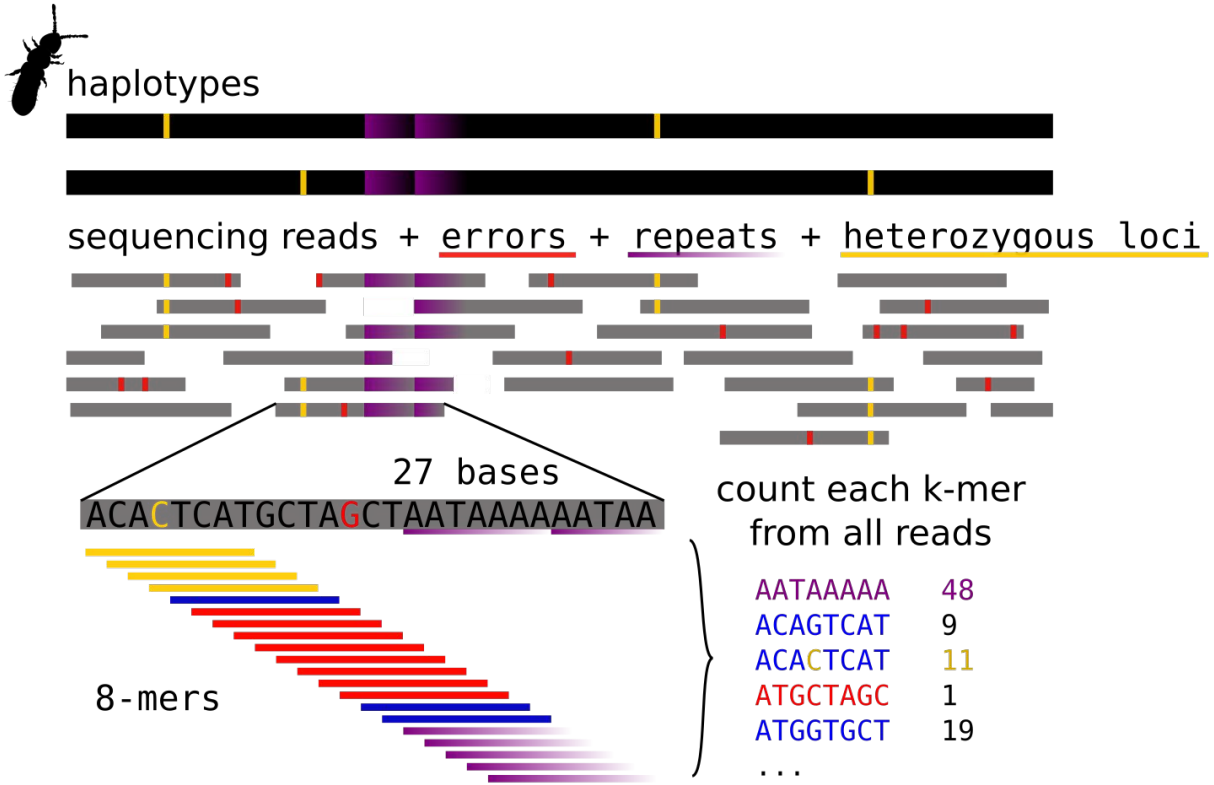
An imperfect and more realistic k-mer histogram



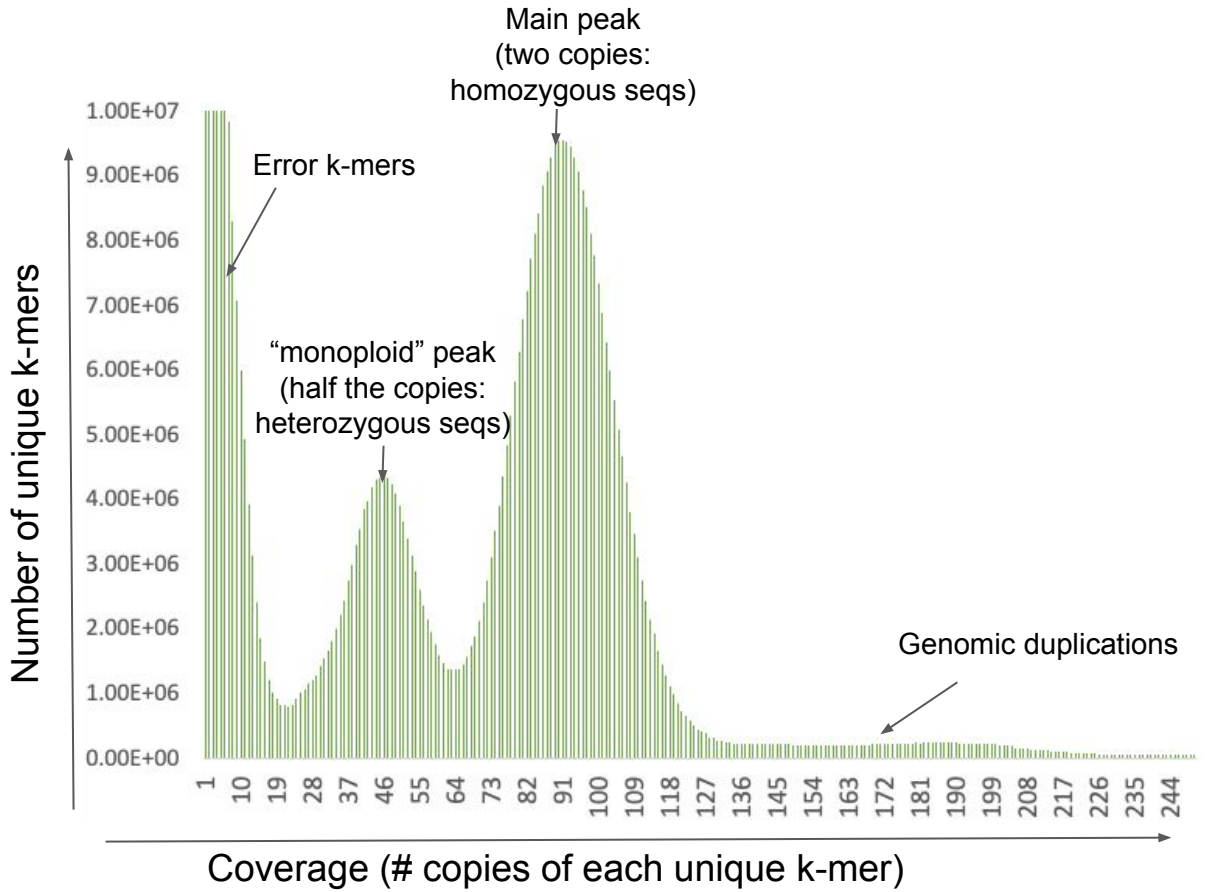
An imperfect and more realistic k-mer histogram



Counting k-mers…

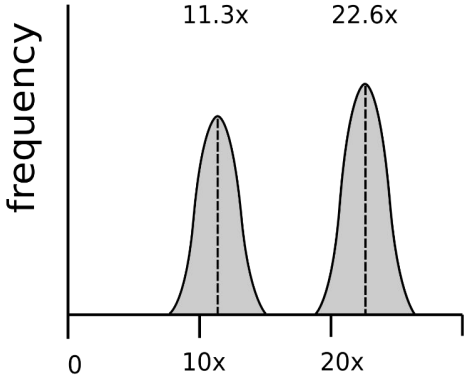


A realistic imperfect and non-inbred k-mer histogram

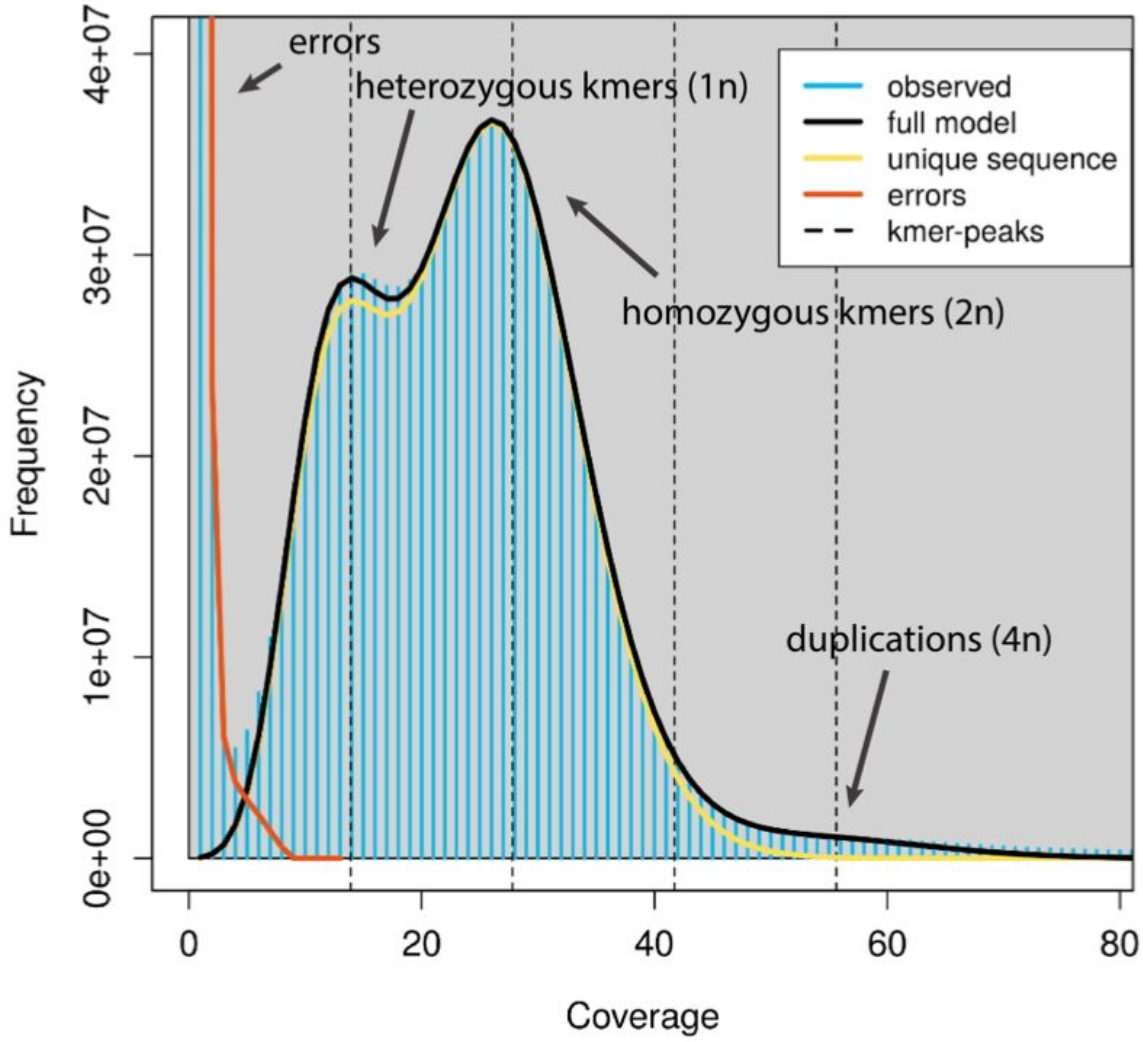


The principle of stoichiometry

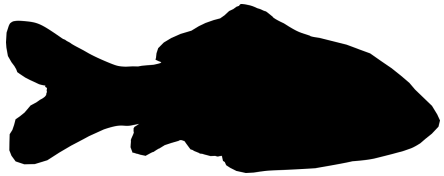
The *k*-mers that occur in a single copy in the genome (e.g. heterozygous in diploid) will have on average ½ coverage of the the *k*-mers that are present in two copies (homozygous in diploid)



There are millions of *k*-mers in each datasets, the deviations are not statistical but technical or biological!!!


errors
heterozygous kmers (1n)
homozygous kmers (2n)
duplications (4n)
observed
full model
unique sequence
errors
kmer-peaks
Frequency
0e+00
1e+07
2e+07
3e+07
4e+07
0
20
40
60
80
Coverage

Common bream



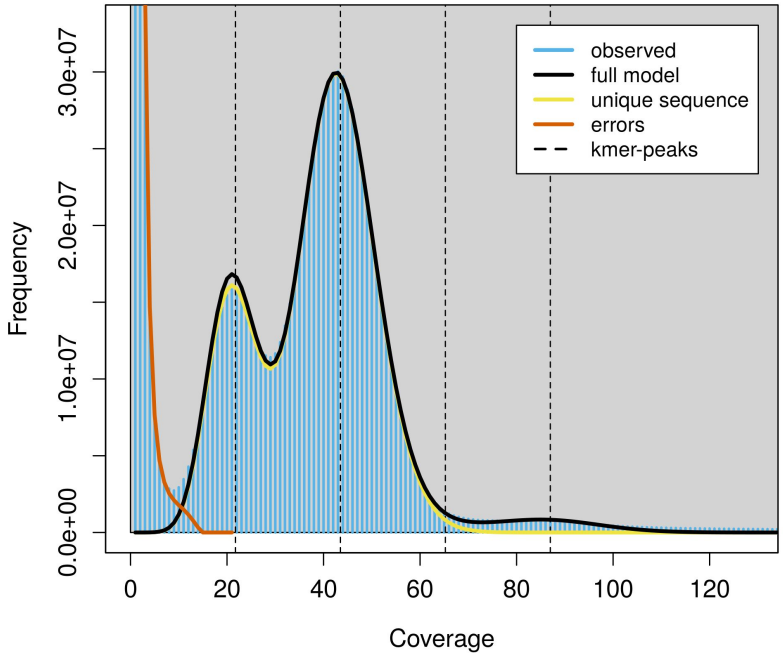
Do we have enough coverage? Did sequencing run go well?

Rule of thumb - do you see clear genomic peak(s)?

Yes: Probably

Kind of: Possibly

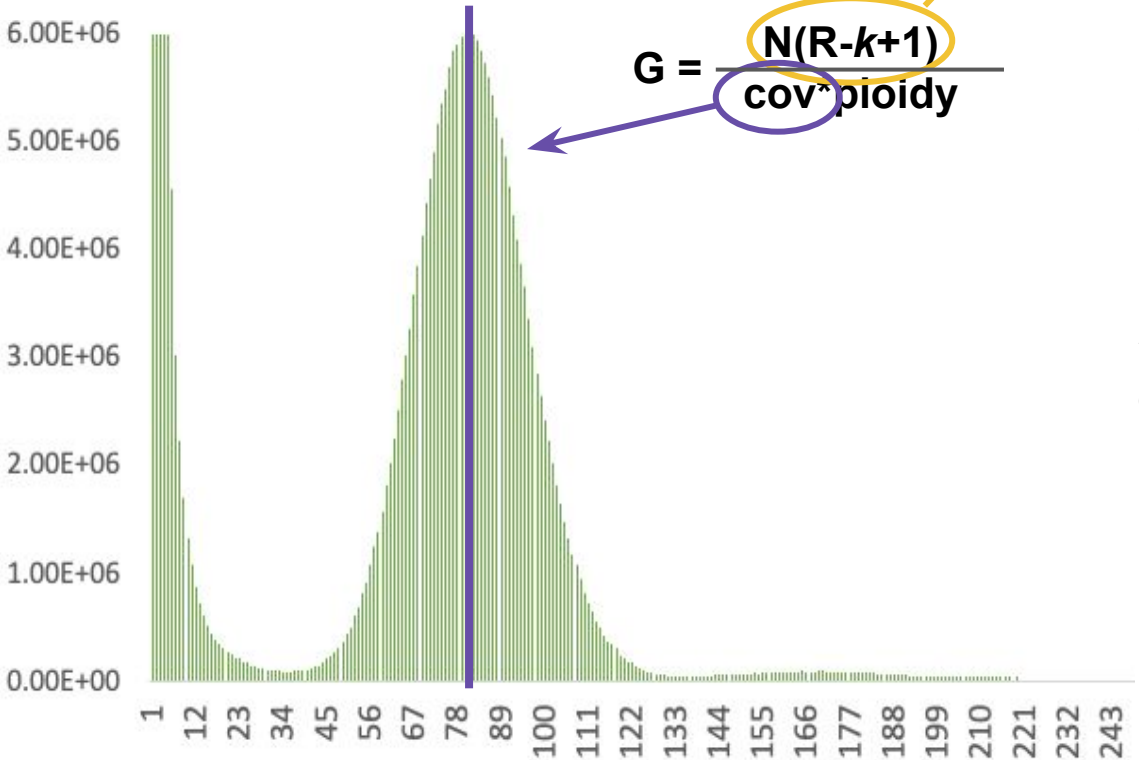
No: Most likely no!



Part 2: Genome profiling



Modeling genome size



$$G = \frac{N(R-k+1)}{cov * ploidy}$$

∑ freq (y-axis) * cov (x-axis)

G = genome size (bp)

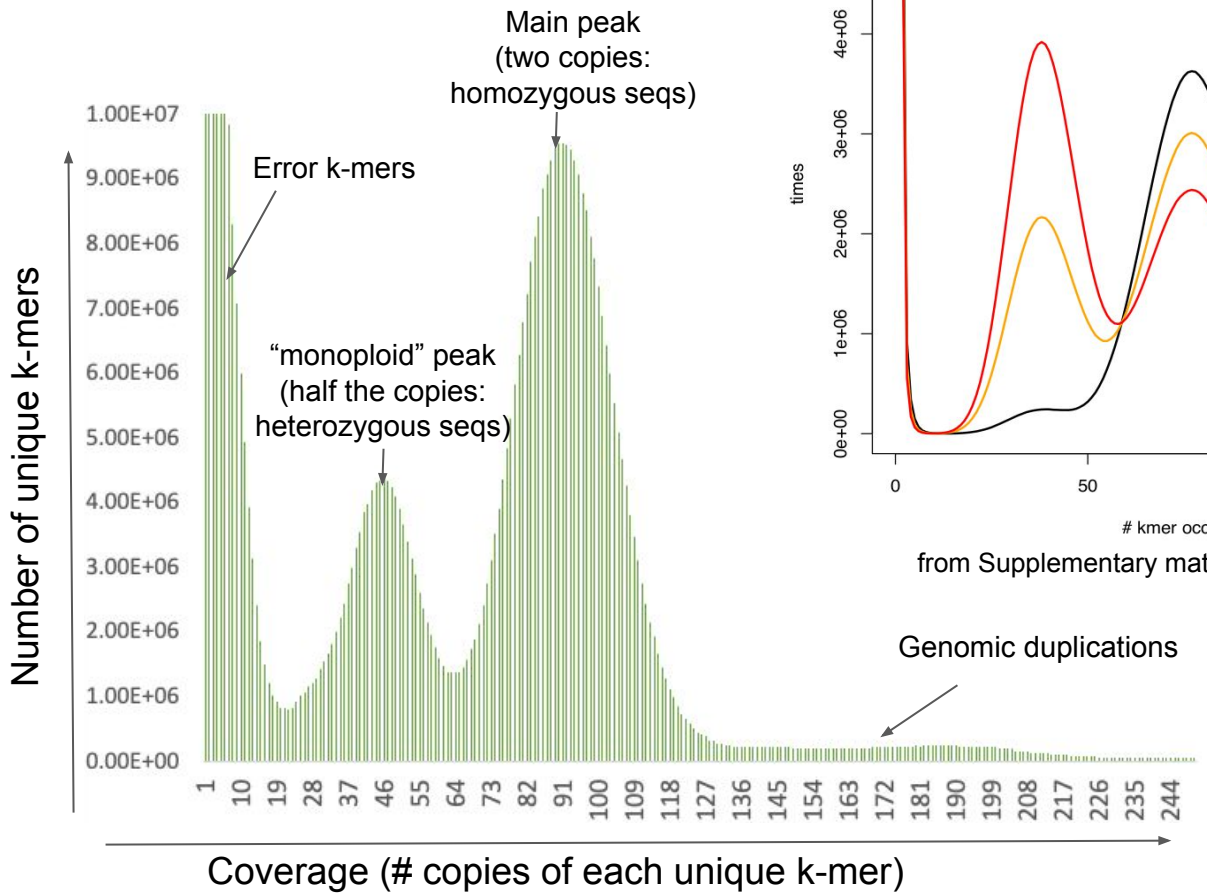
N = number of reads

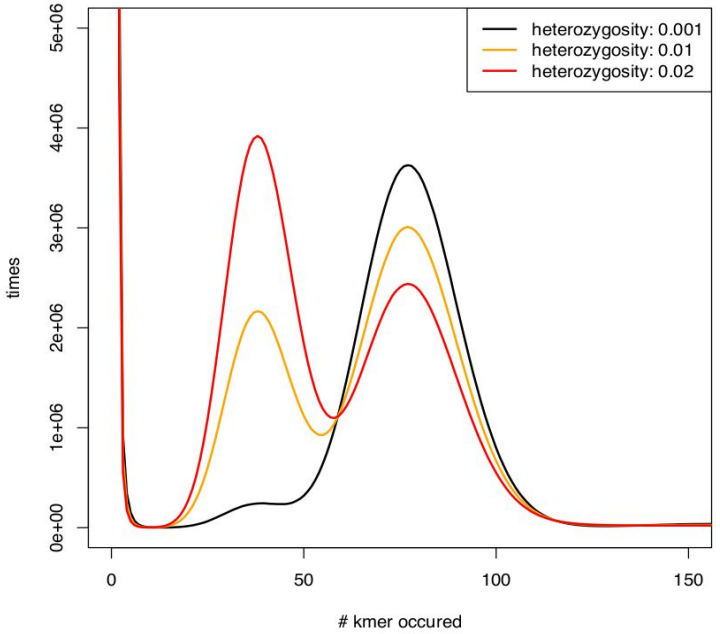
R = read length

k = k-mer length

cov = peak coverage

Modeling heterozygosity





from Supplementary materials of Vurture et al. 2017

The genome profiling workflow

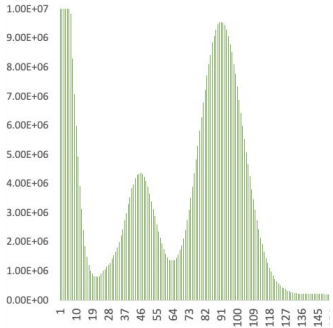
sequencing reads



counting k-mers



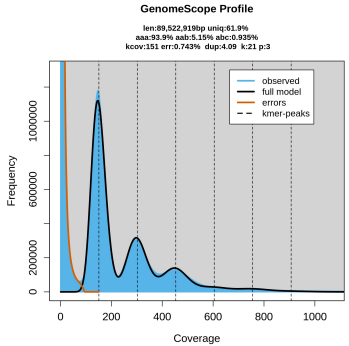
k-mer spectrum



fitting a model



“genome profiling”



k-mer counters:

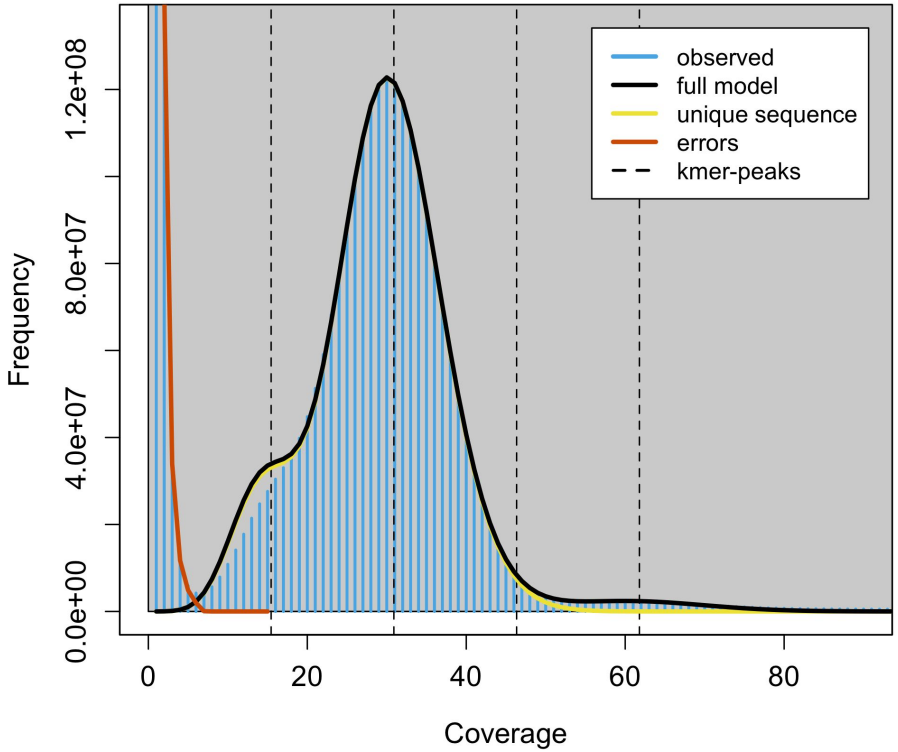
- JellyFish
- KMC
- **FastK**

Genome profiling tools:

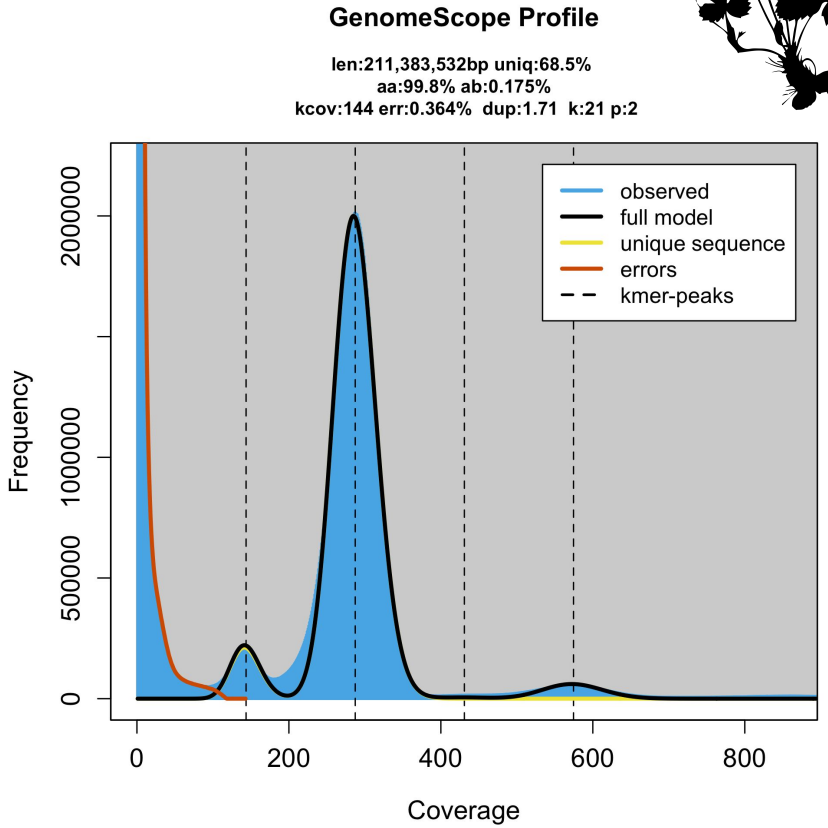
- **GenomeScope**
- RESPECT
- Tetmer

For sane high coverage genomes spectra look good!

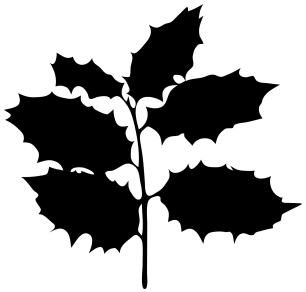








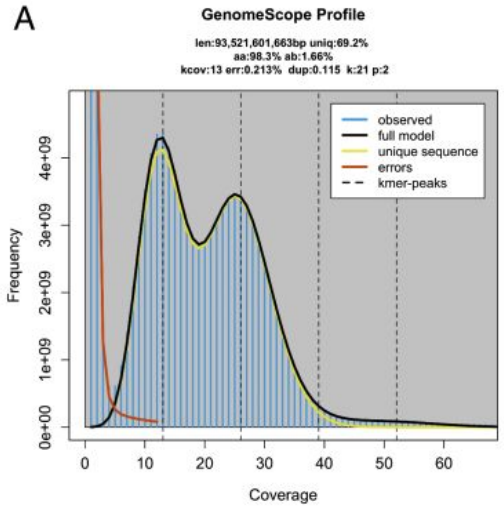
Coverage patterns

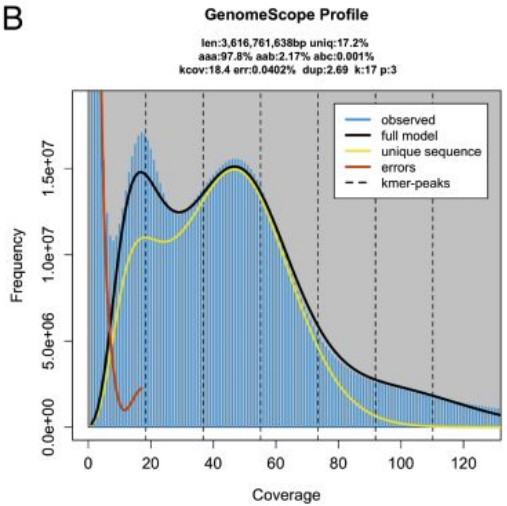


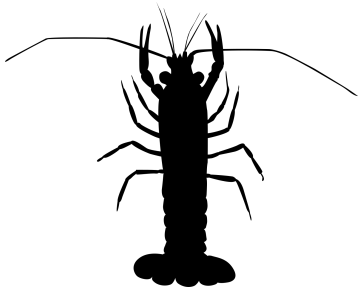
Great



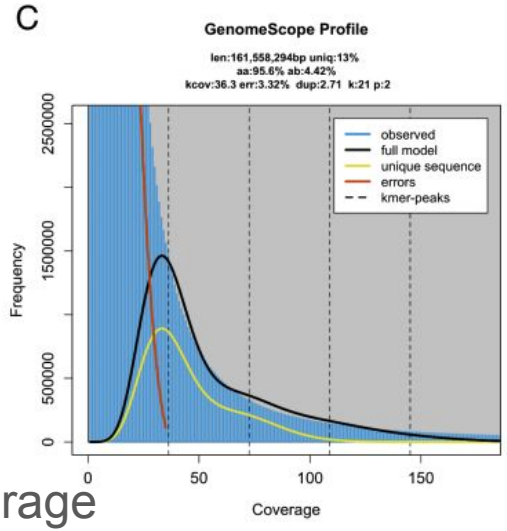
Not enough coverage

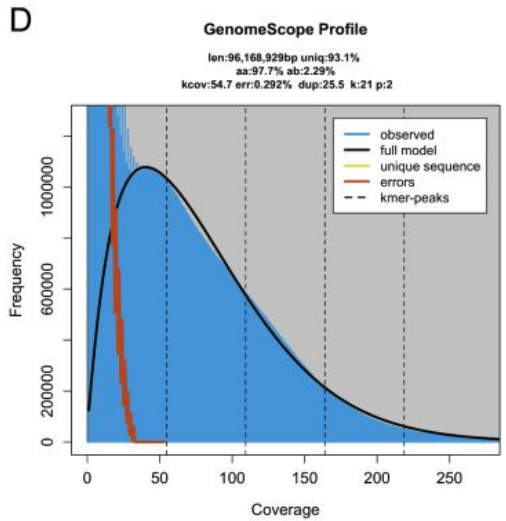


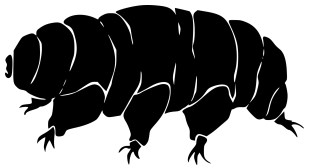




Perhaps ok?







Contamination

Contamination is one and the only exception of possible good run without a clear coverage peak

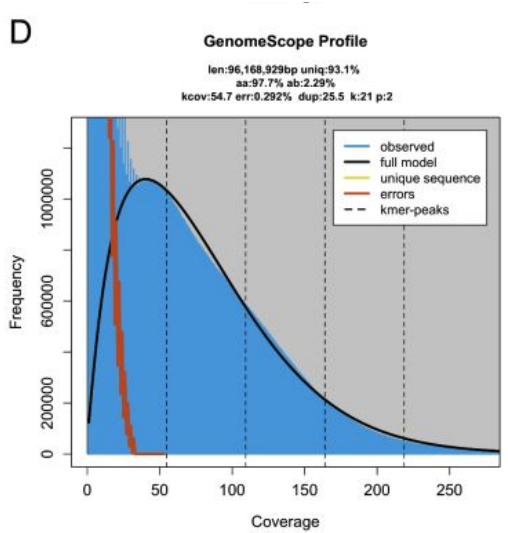
Would apply only for samples where very high levels of contamination can be expected… sequencing guts, environmental or otherwise messy samples…

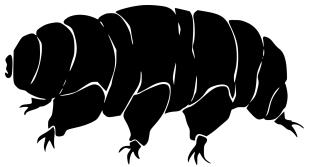
Rule of thumb - do you see clear genomic peak(s)?

No? Is it a tiny messy sample, is the genome size in the right ballpark?

Yes - Might be fine

No - Low coverage!





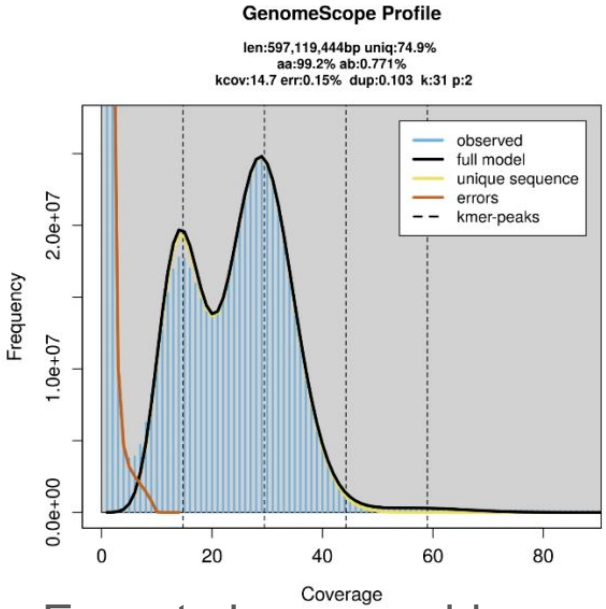
Contamination

Sequencing biases can make “coverage bridge”

(GA-rich regions)

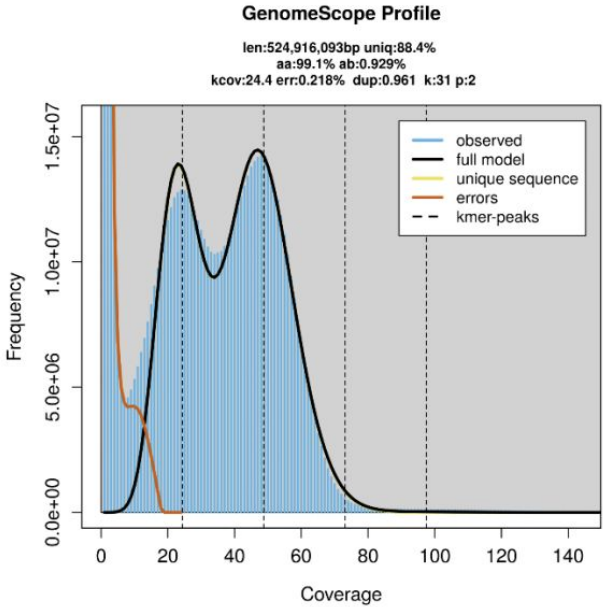


A. stone loach

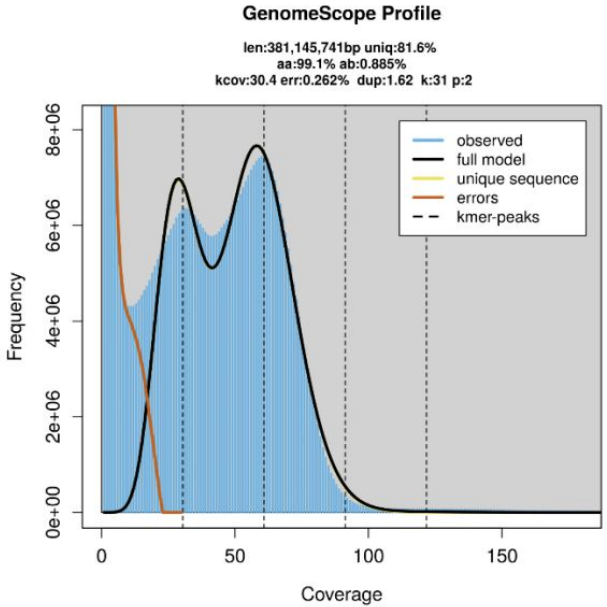


Expect nice assembly

B. European flounder

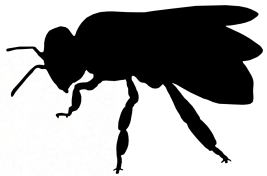


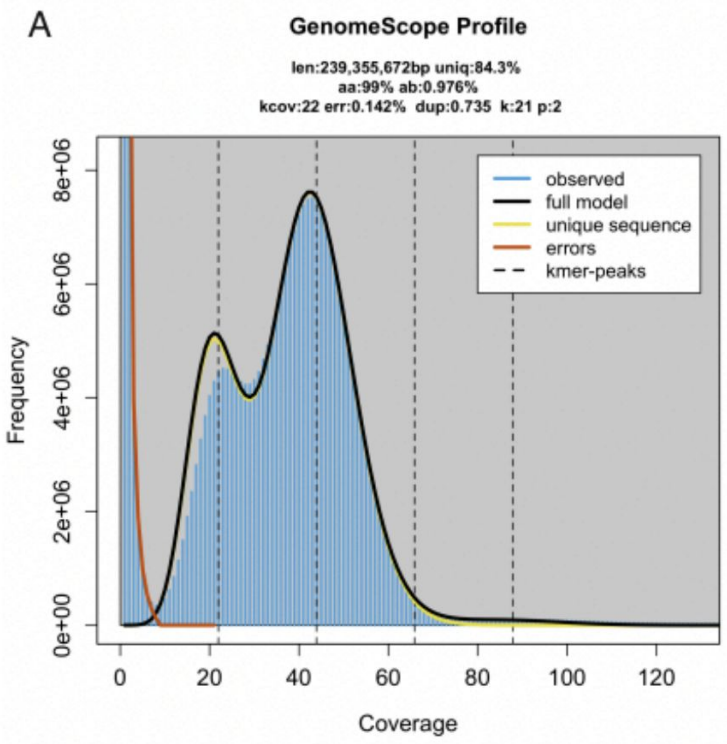
C. nine-spined stickleback

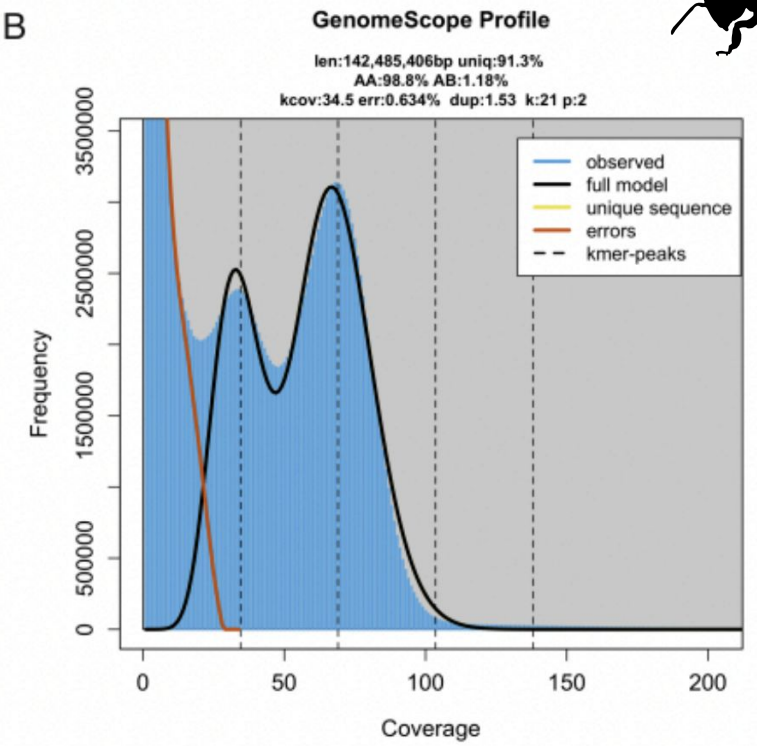


Expect fragmented assembly

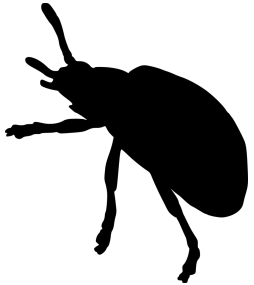
“Coverage bridge” can be due to poor sample handling





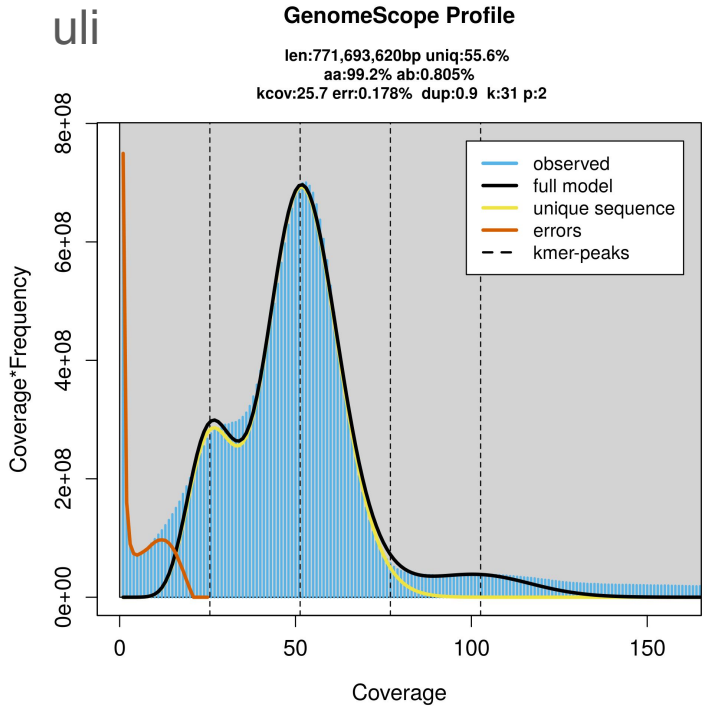


“Coverage bridge” can be due amplification



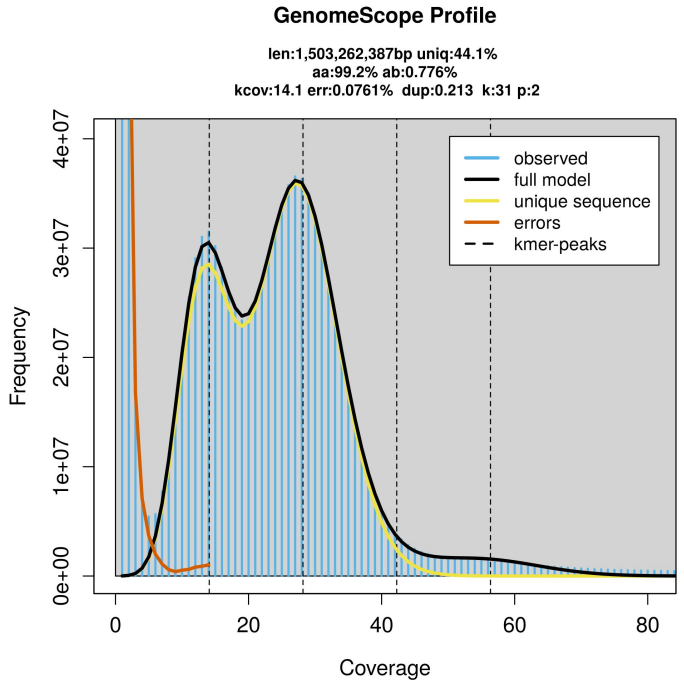
Chrysolina herbacea

uli



Chrysolina graminis

li

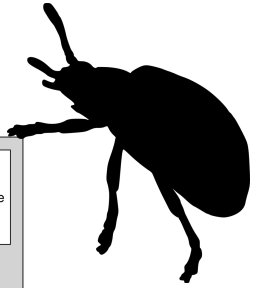


Example by Ksenia Krasheninnikova

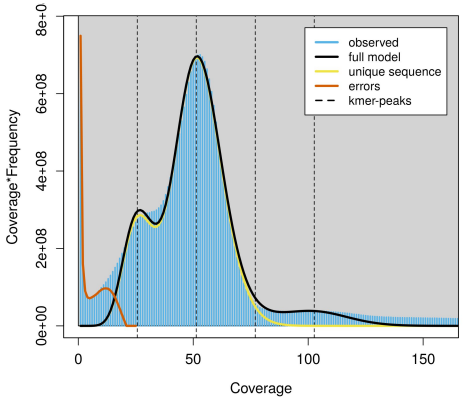
Chrysolina herbacea

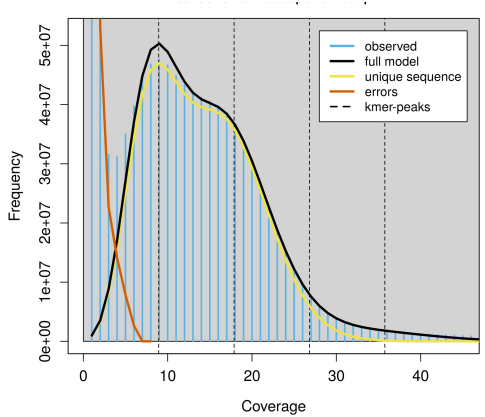
Chrysolina graminis

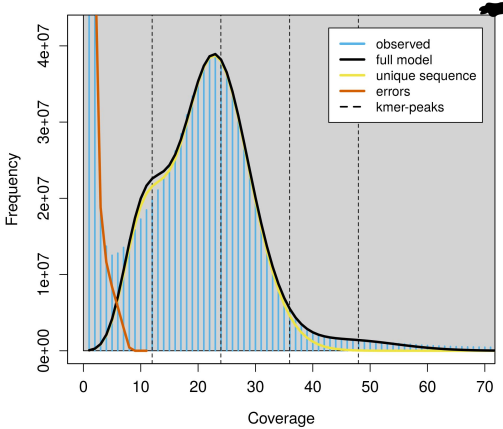
Chrysolina oricalcia



uli

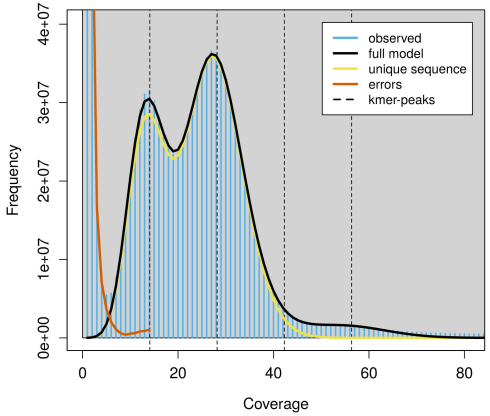


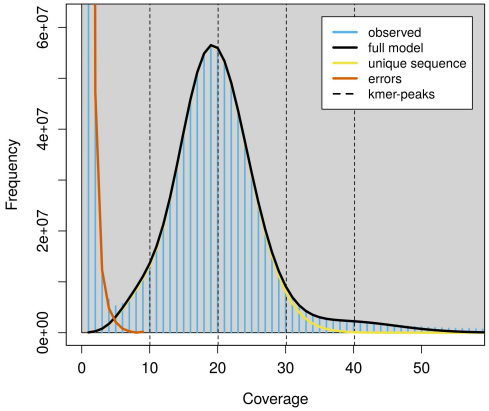




All bridges

li





Ok

Example by Ksenia Krasheninnikova

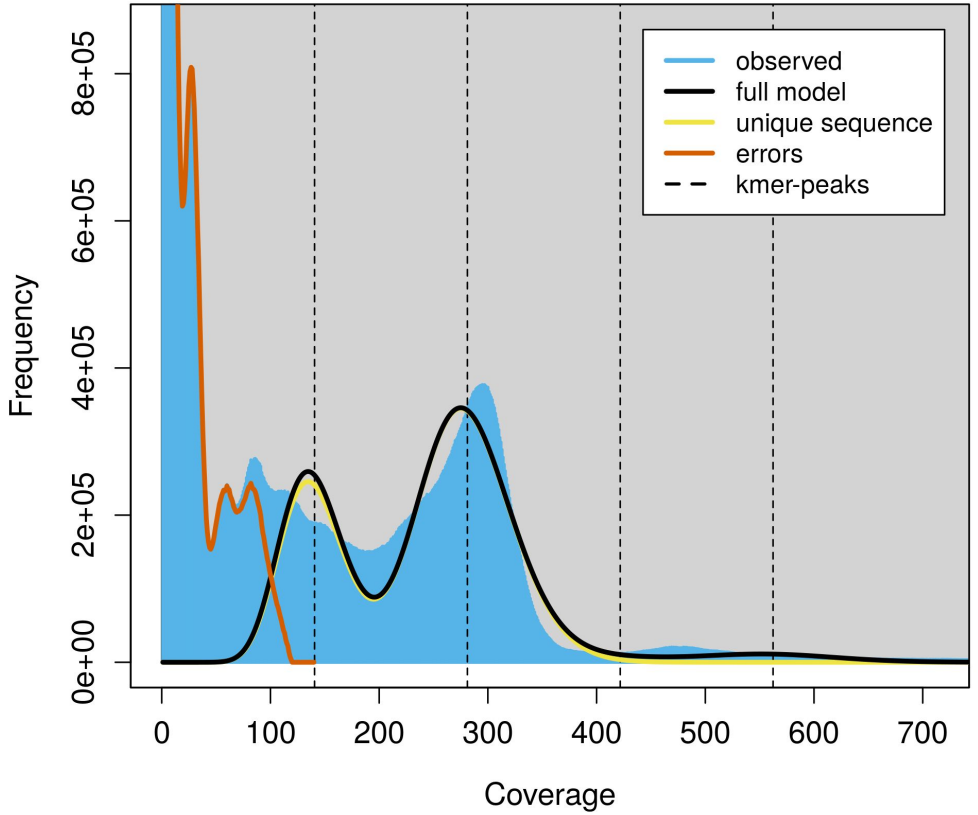
“Pooled” sequencing also make bridges



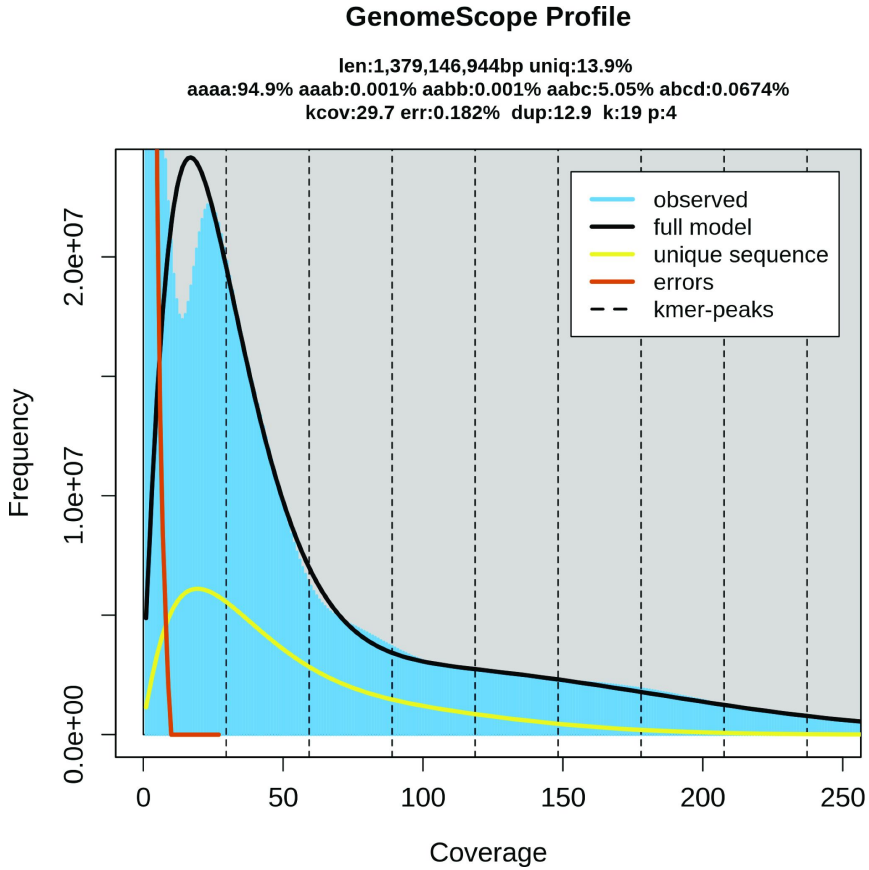
Stylops aterrimus

- One clear genomic peak
- Lots of minor peaks that do not see to have clear stochasticity to the main peak

> looks like pool of genotypes, possibly just because the sequenced female was mated?



What do you think about this spectrum?



10.1371/journal.pone.0322457

Where to find more…

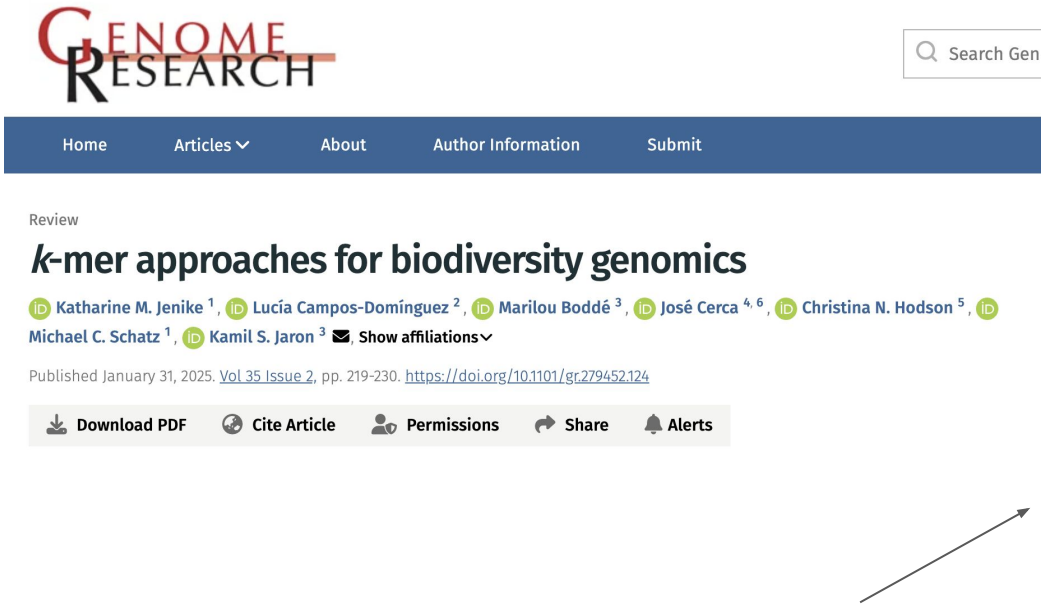


Table of content

Introduction

- Concept of k-mers

k-mer spectra analysis

- Introduction to K-mer spectra analysis
  - Generating k-mer spectra tutorial
- Basics of genome modeling
  - manual model fitting (for better understanding of the underlying model)
  - simple diploid
- Common difficulties in characterisation of diploid genomes using k mer spectra analysis
  - low coverage (pitfall) - to be merged
  - very homozygous diploid
  - highly heterozygous diploid
  - Genome size of a repetitive genome (pitfall)

https://github.com/KamilSJaron/k-mer-approaches-for-biodiversity-genomics/wiki

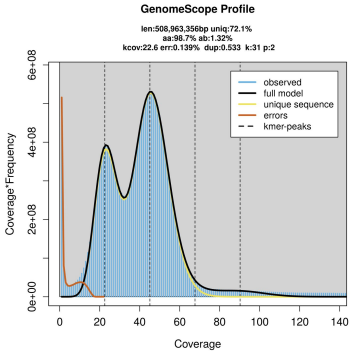
There is a place for all the freaks…

all  data are available on ToLQC

https://tolqc.cog.sanger.ac.uk/



Shane McCarthy



GenomeScope plots

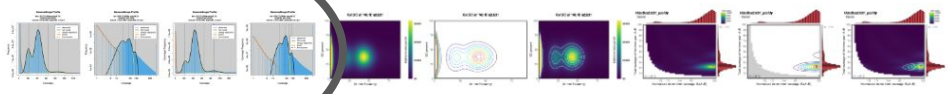
k-mer histogram

```
1       1264991662
2       128715689
3       35706231
4       14690890
5       7658823
6       4750866
7       3423518
8       2858068
9       2737239
10      2944294
11      3473705
12      4288452
13      5379104
14      6746789
15      8374781
16      10152558
17      11869712
18      13391733
19      14663647
20      15457083
21      15833391
22      15731512
23      15224706
24      14487391
25      13626577
```

Search

show all plots | hide all plots

source	specimen	*k*-mer	*k*-cov	haploid size	repeat (%)	heterozygosity (%)	model fit (%)	model error (%)	histogram
pacbio	AbrBra2	31	21.75	1,060,723,998	35.95	0.57	97.39	0.13	histogram.txt



Running GenomeScope on a server

http://genomescope.org/genomescope2.0/

GenomeScope 2.0

Estimate genome heterozygosity, repeat content, and size from sequencing reads using a kmer-based statistical approach.

Run GenomeScope

Click or drop .histo file here to upload

Description	my sample
K-mer length	21
Ploidy	2
Max k-mer coverage	-1
Average k-mer coverage for polyploid genome	-1

Submit

Instructions

Upload results from running Jellyfish or KMC. Example: inputk21.hist

Instructions for running Jellyfish:

1. Download and install jellyfish from: http://www.genome.umd.edu/jellyfish.html#Release
2. Count k-mers using jellyfish:

```
$ jellyfish count -C -m 21 -s 1000000000 -t 10 *.fastq -o reads.jf
```

Note you should adjust the memory (-s) and threads (-t) parameters according to your server. This example will use 10 threads and 1GB of RAM. The k-mer length (-m) may need to be scaled if you have low coverage or a high error rate. You should always use "canonical k-mers" (-C).

3. Export the k-mer count histogram

```
$ jellyfish histo -t 10 reads.jf > reads.histo
```

Again the thread count (-t) should be scaled according to your server.

4. Upload reads.histo to GenomeScope

Instructions for running KMC:

1. Download and install KMC from: http://sun.aei.polsl.pl/REFRESH/index.php?page=projects&project=kmc&subpage=download
2. Count k-mers using KMC:

```
$ mkdir tmp
```

```
$ ls *.fastq > FILES
```

```
$ kmc -k21 -t10 -m64 -ci1 -cs10000 @FILES reads tmp/
```

Note you should adjust the memory (-m) and threads (-t) parameters according to your server. This example will use 10 threads and 64GB of RAM. The k-mer length (-k) may need to be scaled if you have low coverage or a high error rate. The lower (-ci) and upper (-cs) bounds exclude k-mers with counts outside these boundaries. FILES is a file with a list of input files.

3. Export the k-mer count histogram

k-mer histogram

1	1264991662
2	128715689
3	35706231
4	14690890
5	7658823
6	4750866
7	3423518
8	2858068
9	2737239
10	2944294
11	3473705
12	4288452
13	5379104
14	6746789
15	8374781
16	10152558
17	11869712
18	13391733
19	14663647
20	15457083
21	15833391
22	15731512
23	15224706
24	14487391
25	13626577



Mike Schatz

Running GenomeScope in terminal

```
[kj11@mib117963s:~/Projects/kmer_stuff/genomescope2.0$ genomescope.R --help
usage: /Users/kj11/bin/genomescope.R [-h] [-v] [-i INPUT] [-o OUTPUT]
                                     [-p PLOIDY] [-k KMER_LENGTH]
                                     [-n NAME_PREFIX] [-l LAMBDA]
                                     [-m MAX_KMERCOV] [--verbose]
                                     [--no_unique_sequence] [-t TOPOLOGY]
                                     [--initial_repetitiveness INITIAL_REPETITIVENESS]
                                     [--initial_heterozygosities INITIAL_HETEROZYGOSITIES]
                                     [--transform_exp TRANSFORM_EXP]
                                     [--testing] [--true_params TRUE_PARAMS]
                                     [--trace_flag] [--num_rounds NUM_ROUNDS]
                                     [--fitted_hist]
                                     [--start_shift START_SHIFT]
                                     [--typical_error TYPICAL_ERROR]
```

https://github.com/tbenavi1/genomescope2.0

The workflow for this morning



FastK

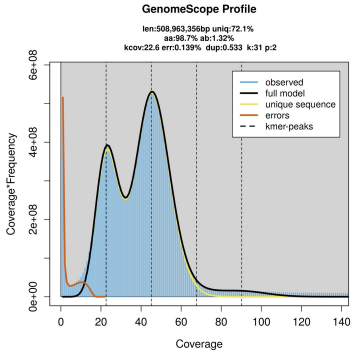
creates a *k*-mer database and

kmer	coverage
AAAAAAAAAACAACGT	28
AAAAATAACACAACGT	31
AAAAATAACACAACGG	1
AAAAATAACGCAACGT	47
AAAATTTACGCAACGT	2
AAAATTTACGCAACGA	17
...	...

k-mer histogram

1	1264991662
2	128715689
3	35706231
4	14690890
5	7658823
6	4750866
7	3423518
8	2858068
9	2737239
10	2944294
11	3473705
12	4288452
13	5379104
14	6746789
15	8374781
16	10152558
17	11869712
18	13391733
19	14663647
20	15457083
21	15833391
22	15731512
23	15224706
24	14487391
25	13626577

GenomeScope



Your turn…

https://tinyurl.com/Scalable-genomics-Module1

- Build k-mer spectrum
- Fit example GenomeScope model
- Fit many more
  - discuss what do you see / think !!!

Part 3: Dissection of GenomeScope model



The idea behind GenomeScope genome model

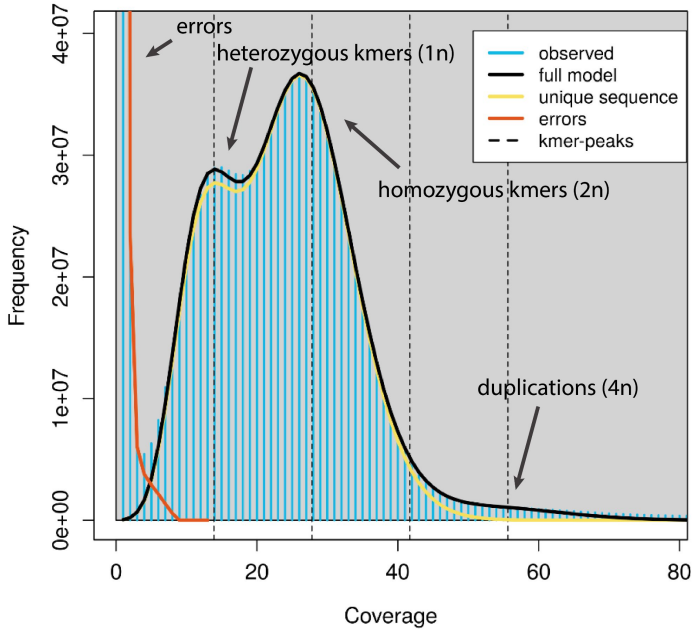
```
frequency ~ error peak +

              1n peak +

              2n peak +

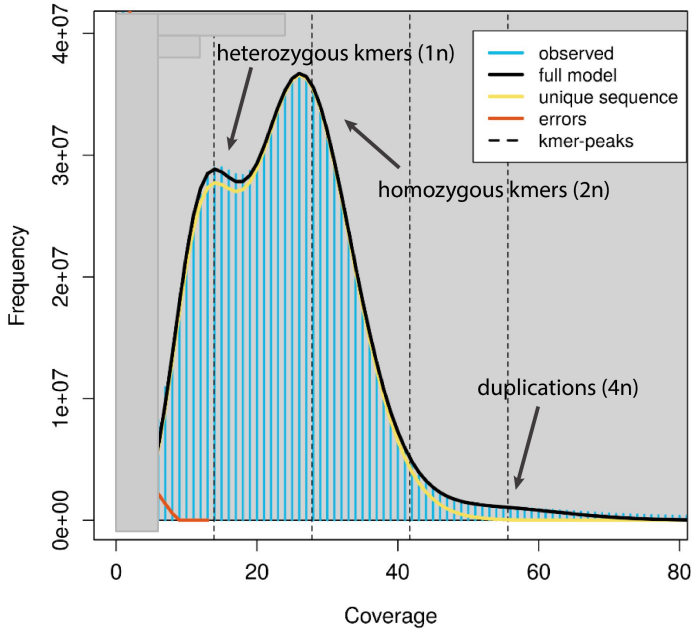
              3n peak +

              4n peak
```



GenomeScope “error” model

```
frequency ~ 1n peak +
            2n peak +
            3n peak +
            4n peak
```

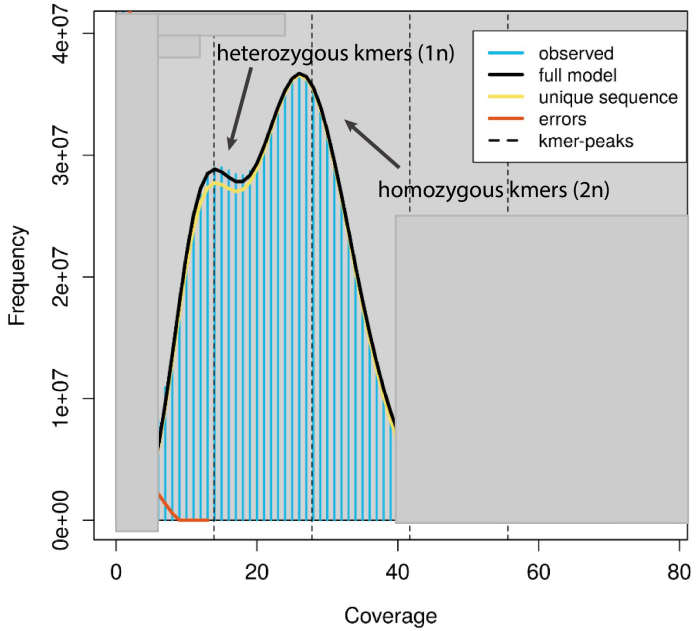


GenomeScope actually does NOT explicitly model errors
They are modelled as left-side residuals

Simplified GenomeScope genome model

```
frequency ~ 1n peak +

            2n peak
```



Modeling coverage as negative binomial

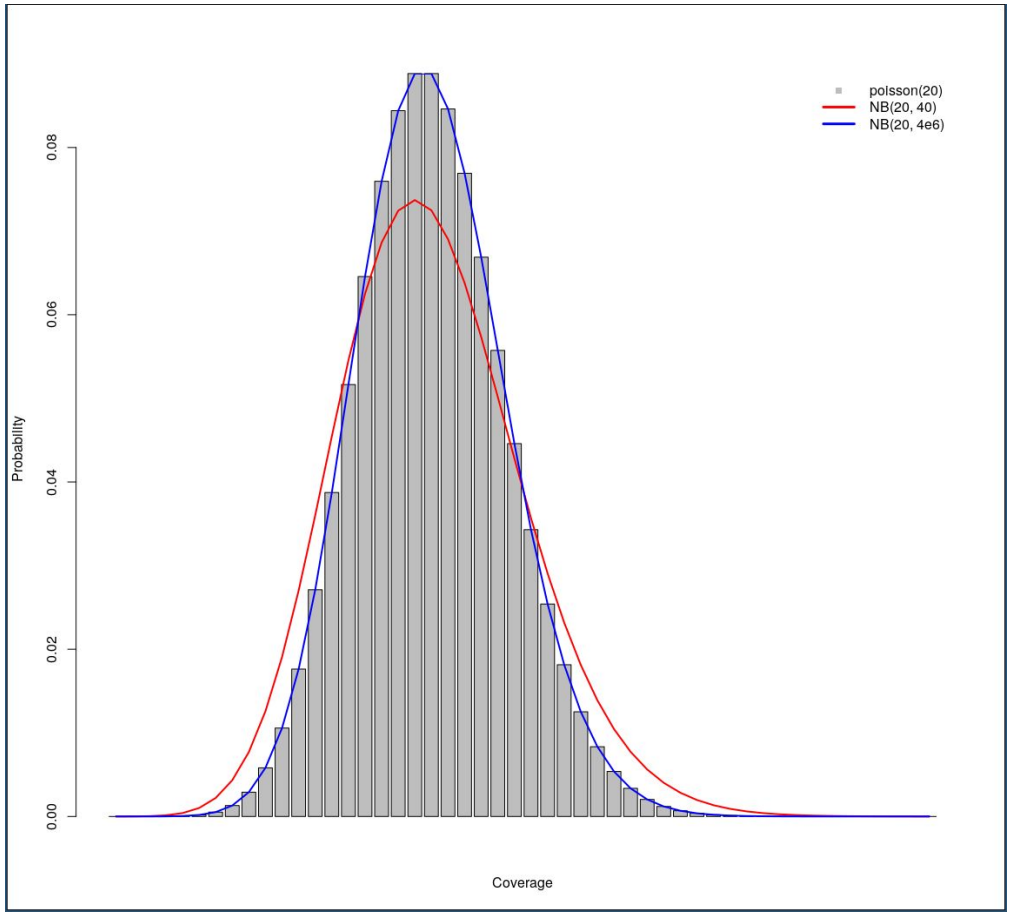
Poisson (λ) ~

$$NB(\lambda, \frac{\lambda}{\xi}) \text{ for } \xi \to 0$$

ξ - overdispersion parameter

There is a good reason why I express it as this moment as

$$\frac{\lambda}{\xi}$$



GenomeScope genome model - simplified

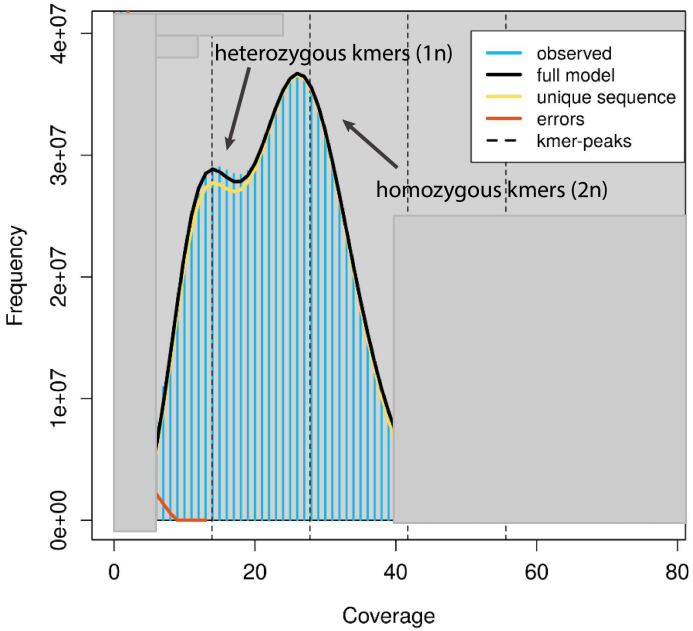
$$\text{Probability} \sim \alpha\, NB(\ kcov,\ \ kcov\ /\ \xi) + \beta\, NB(2*kcov,\ 2*kcov\ /\ \xi)$$

α and **β** are relative contributions of 1n and 2n peaks

So **α + β = 1**

kcov is 1n coverage and

ξ is the overdispersion parameter from the last slide



Modeling coverage as negative binomial

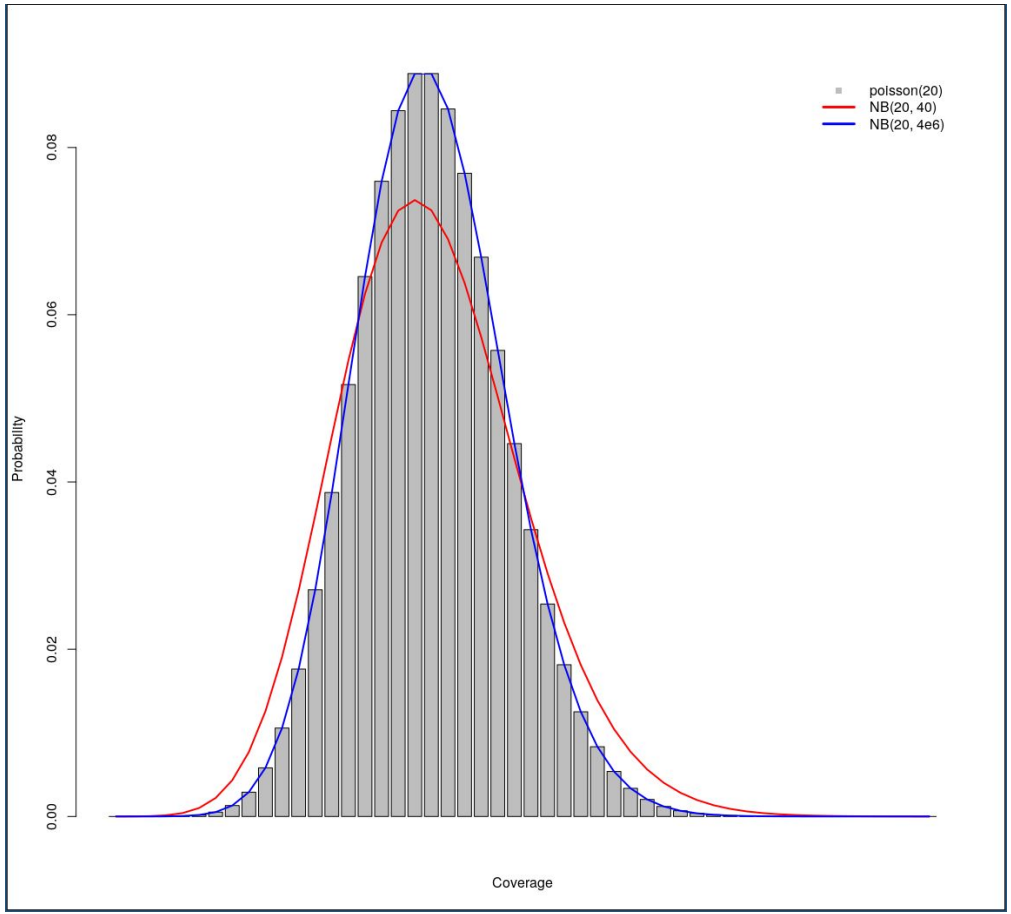
Poisson (λ) ->

$$NB(\lambda, \frac{\lambda}{\xi})$$ for ξ -> 0

ξ - overdispersion parameter

$$\sum NB(\lambda, \frac{\lambda}{\xi}) = 1$$

Each PROBABILITY distribution sums to 1

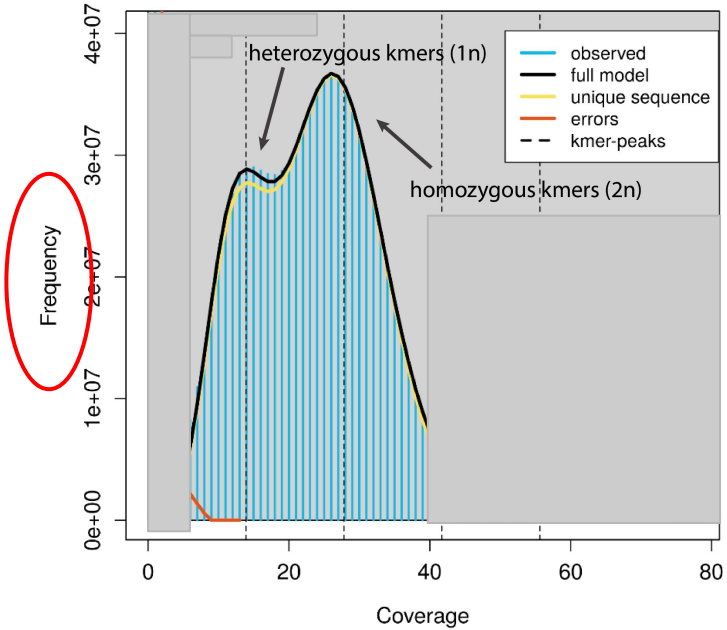


We need a factor to scale the model to k-mer frequencies

```
Frequency  ~ (α NB(  kcov,   kcov / ξ)  +
              β NB(2*kcov, 2*kcov / ξ))
```

```
Probability != Frequency

  -> we need a factor that turns Prob -> Freq
```



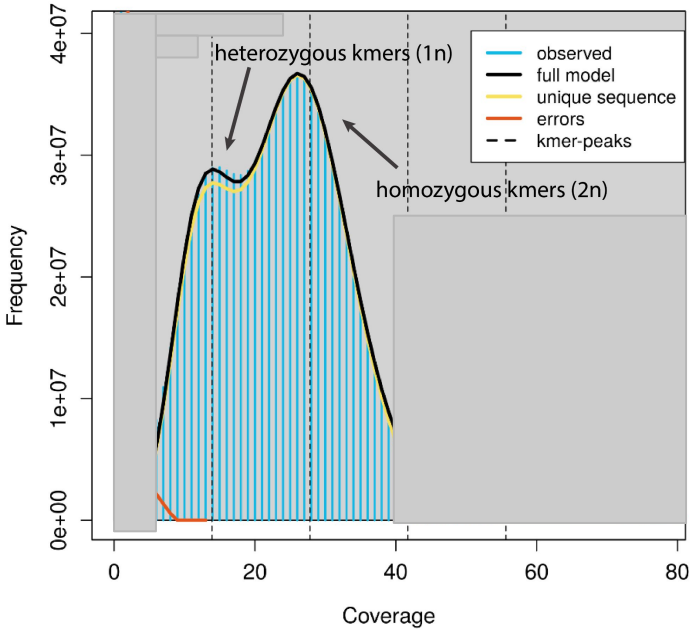
Working GenomeScope genome model

$$Frequency \sim (\alpha\ NB(kcov, kcov / \xi) + \beta\ NB(2*kcov, 2*kcov / \xi)) * length$$

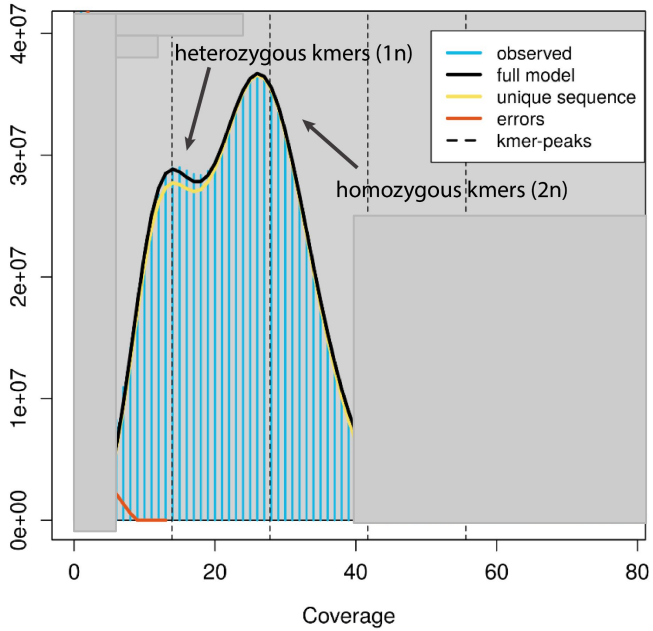
α and **β** are relative contributions of 1n and 2n peaks

kcov is 1n coverage and bias is overdispersion parameter

length is a factor that will scale the probability distribution (and moreless corresponds to the haploid genome size)



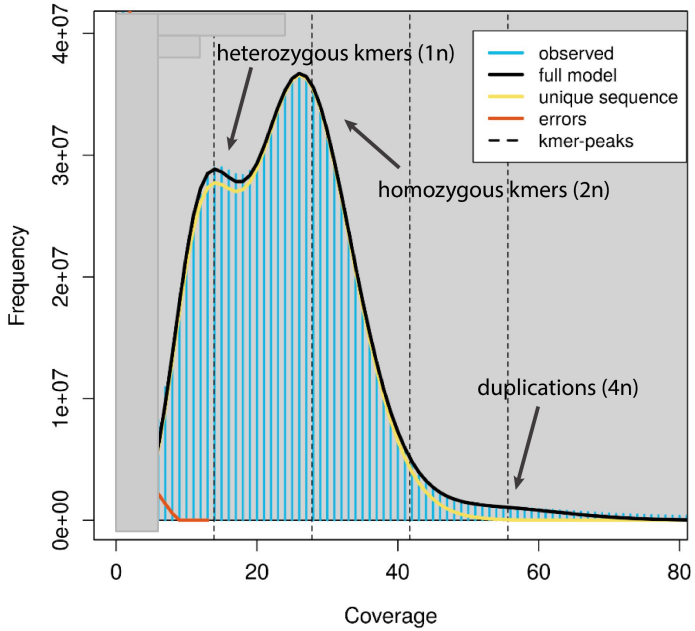
Why moreless? And how to make it right



length is a factor that will scale the probability distribution (and moreless corresponds to the haploid genome size)

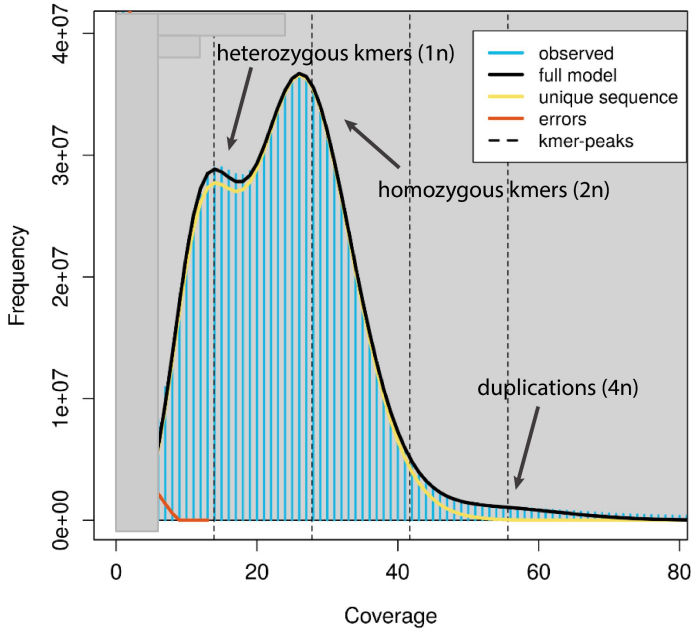
We ignore every repetition in the genome!

and they represent a large portion of our genomes!



Calculating genome size

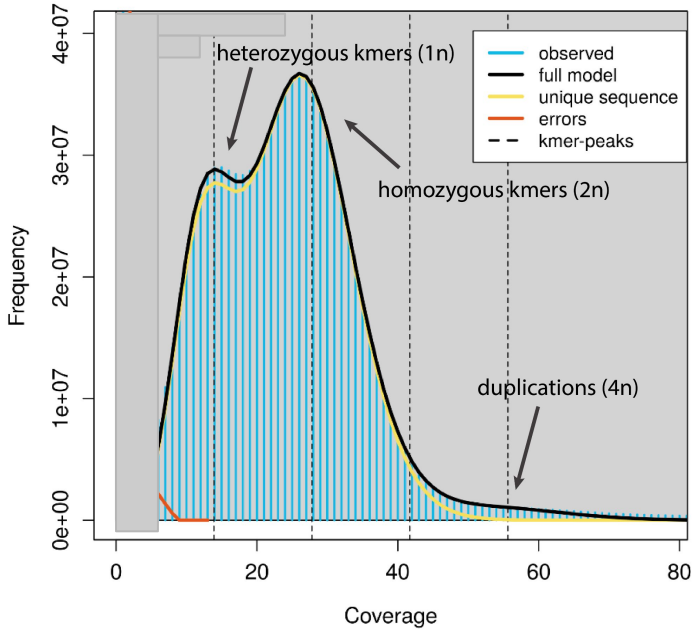
$$coverage = \frac{sequencing\ yield}{genome\ size}$$



Calculating genome size

$$coverage = \frac{sequencing\ yield}{genome\ size}$$

$$genome\ size = \frac{sequencing\ yield}{coverage}$$

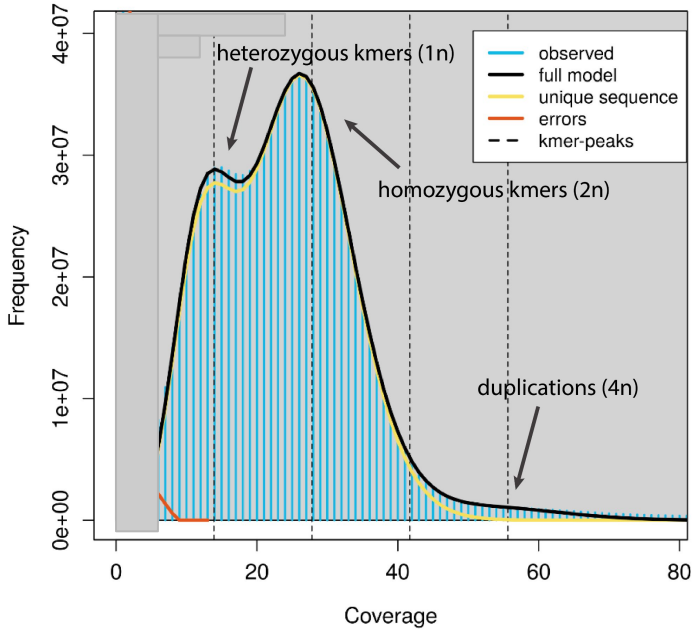


Calculating genome size

$$coverage = \frac{sequencing\ yield}{genome\ size}$$

$$genome\ size = \frac{sequencing\ yield}{coverage}$$

$$genome\ size = \frac{sequencing\ \text{“}k\text{-mer”}\ yield}{k\text{-mer}\ coverage\ *\ ploidy}$$

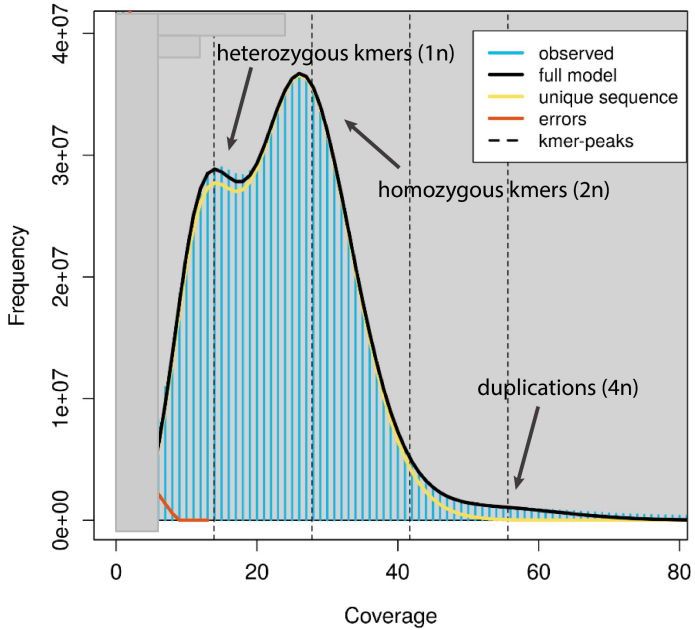


Calculating genome size

"sum all the non-error k-mer coverages"

in k-mer histogram world: *sum(kmer_cov * kmer_freq)*

$$\text{genome size} = \frac{\text{sequencing "k-mer" yield}}{\text{k-mer coverage * ploidy}}$$



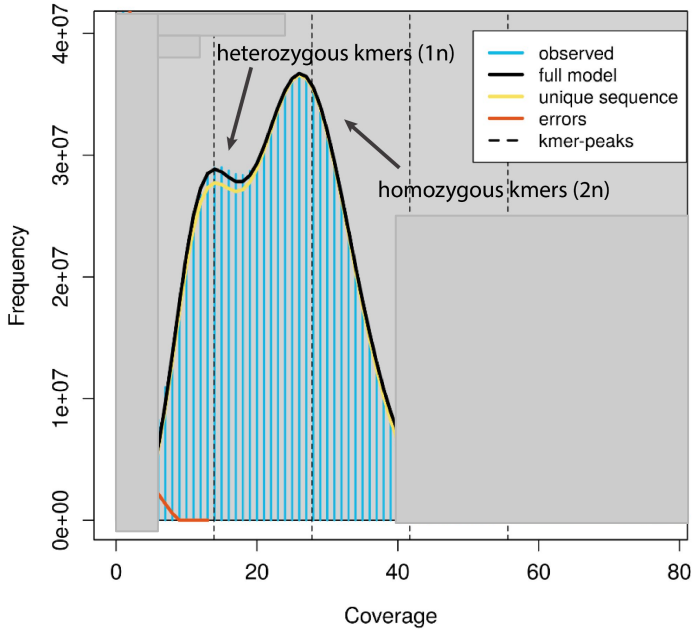
Working GenomeScope genome model

$$\text{Frequency} \sim (\alpha\, NB(\text{kcov}, \text{kcov} / \xi) + \beta\, NB(2*\text{kcov}, 2*\text{kcov} / \xi)) * \text{length}$$

α and **β** are relative contributions of 1n and 2n peaks

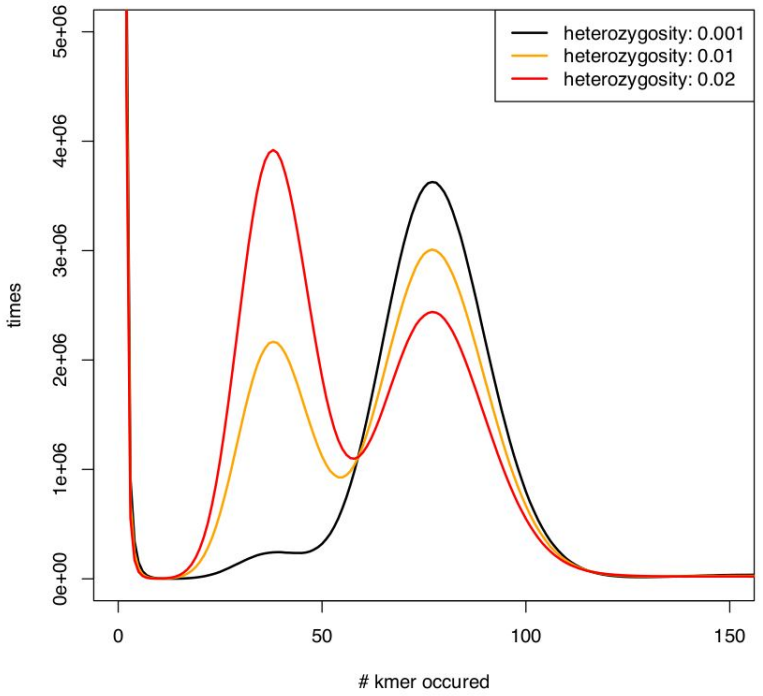
kcov is 1n coverage and bias is overdispersion parameter

length is a factor that will scale the probability distribution (and moreless corresponds to the haploid genome size)



α and β somehow represent heterozygosity

```
Frequency  ~ (α NB(  kcov,   kcov / ξ)  +
              β NB(2*kcov, 2*kcov / ξ))
              * length
```



from Supplementary materials of Vurture et al. 2017

Expressing **α** and **β** as function of heterozygosity

r ⇨ probability of a heterozygous nucleotide (i.e. heterozygosity)

k ⇨ k-mer size

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$(1 - r)^k$ ⇨ prob of homozygous (2n) k-mer

$1 - (1 - r)^k$ ⇨ prob of heterozygous (1n) k-mer

⇨ Each heterozygous site generate 2 het k-mers

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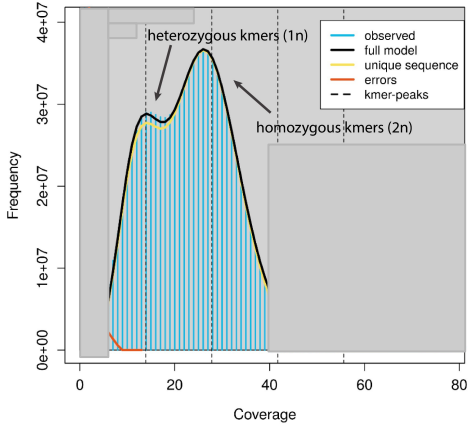
⇨ Each heterozygous site generate 2 het k-mers

$$\alpha = 2 * (1 - (1 - r)^k)$$

$$\beta = (1 - r)^k$$

Working GenomeScope genome model in R

```
Frequency ~ (α NB(  kcov,   kcov / ξ)  +
             β NB(2*kcov, 2*kcov / ξ)) * length
```

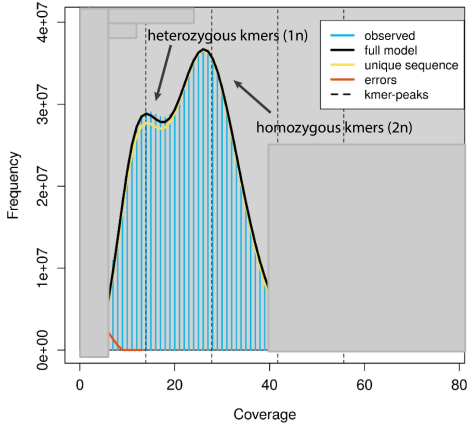


In R:

```
nls(y ~ ((2*(1-(1-r)^k)) * dnbinom(x, size = kcov   / overdispersion, mu = kcov   ) +
         ((1-r)^k)       * dnbinom(x, size = kcov*2 / overdispersion, mu = kcov*2)) * length),
    ...)
```

Working GenomeScope genome model in R

```
Frequency ~ (α NB(  kcov,   kcov / ξ)  +
             β NB(2*kcov, 2*kcov / ξ)) * length
```



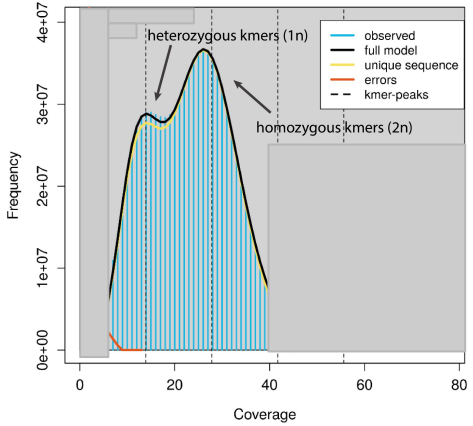
In R:

```
nls(y ~ ((2*(1-(1-r)^k)) * dnbinom(x, size = kcov   / overdispersion, mu = kcov   ) +
         ((1-r)^k)       * dnbinom(x, size = kcov*2 / overdispersion, mu = kcov*2)) * length),
    start = list(r = 0, kcov = 50, overdispersion = 0.5, length = 300e6),
    ...)
```

starting conditions

Working GenomeScope genome model in R

$$Frequency \sim (\alpha\ NB(kcov, kcov / \xi) + \beta\ NB(2*kcov, 2*kcov / \xi)) * length$$



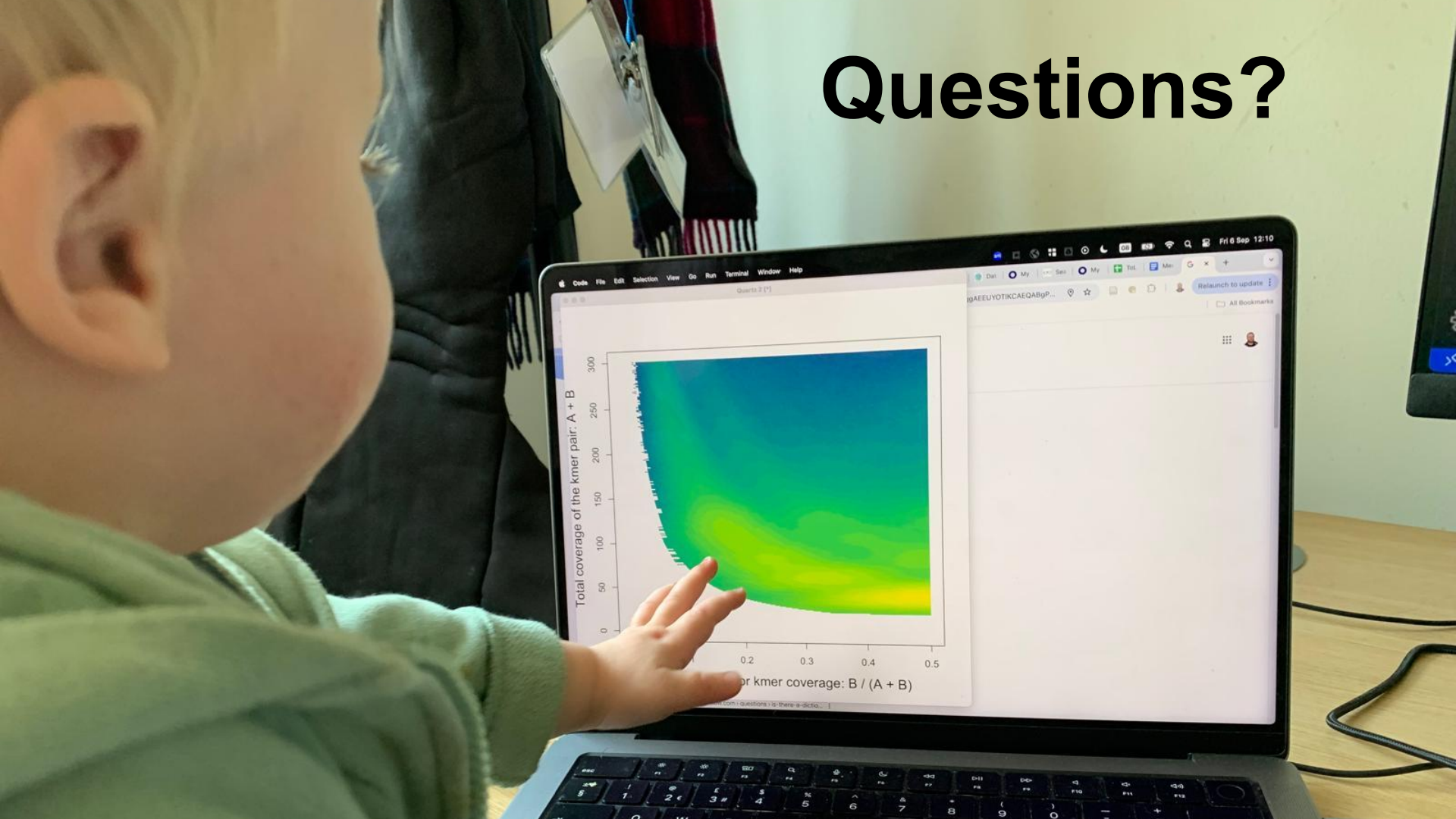
In R:

```
nls(y ~ ((2*(1-(1-r)^k)) * dnbinom(x, size = kcov    / overdispersion, mu = kcov    ) +
         ((1-r)^k)       * dnbinom(x, size = kcov*2 / overdispersion, mu = kcov*2)) * length),
    start = list(r = 0, kcov = 50, overdispersion = 0.5, length = 300e6),
    control = list(minFactor=1e-12, maxiter=40))
```

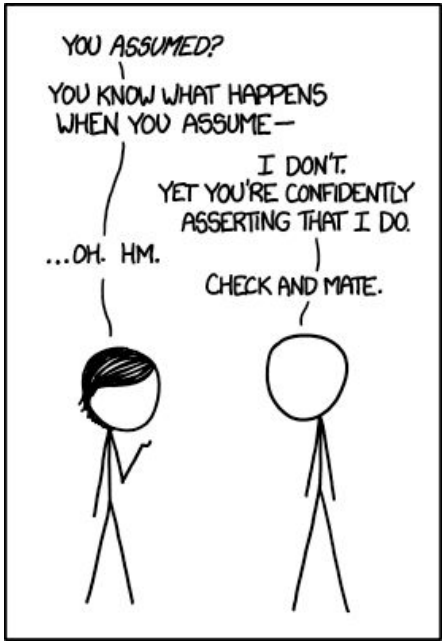
starting conditions

setting for the iterative least square search

Questions?

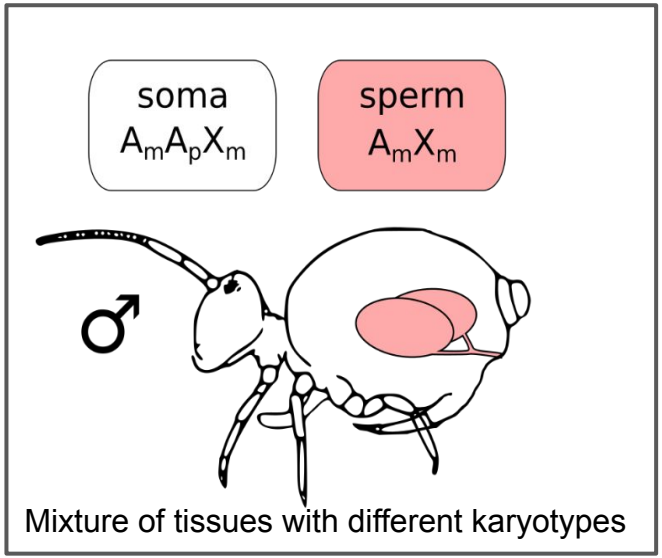

Total coverage of the kmer pair: A + B
0
50
100
150
200
250
300
0.2
0.3
0.4
0.5
r kmer coverage: B / (A + B)

Bonus part: Violating GenomeScope assumptions



Conditions naturally violating GenomeScope assumptions



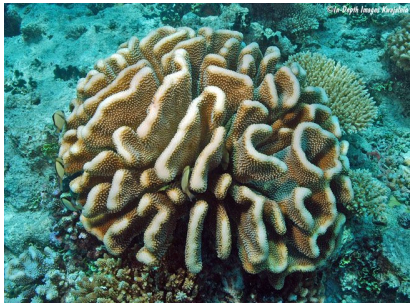




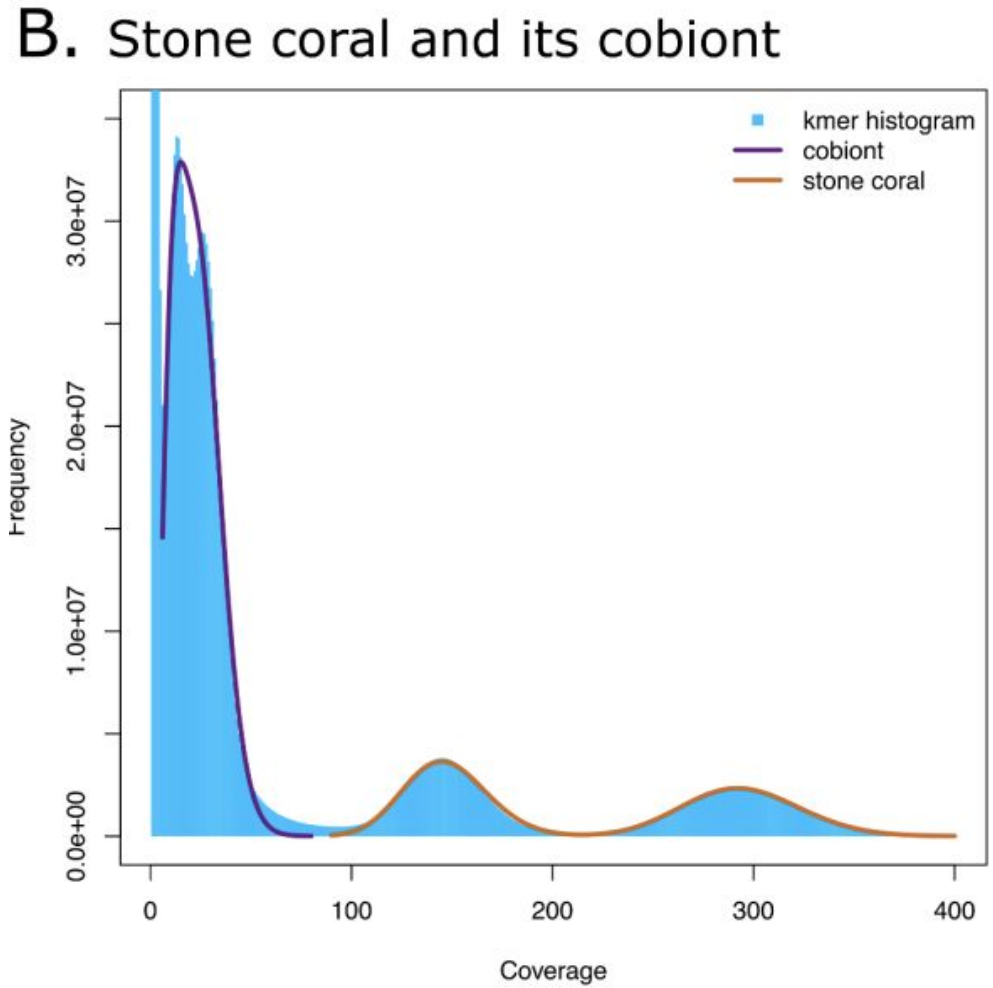
Estimating coverage and genome size of cobionts



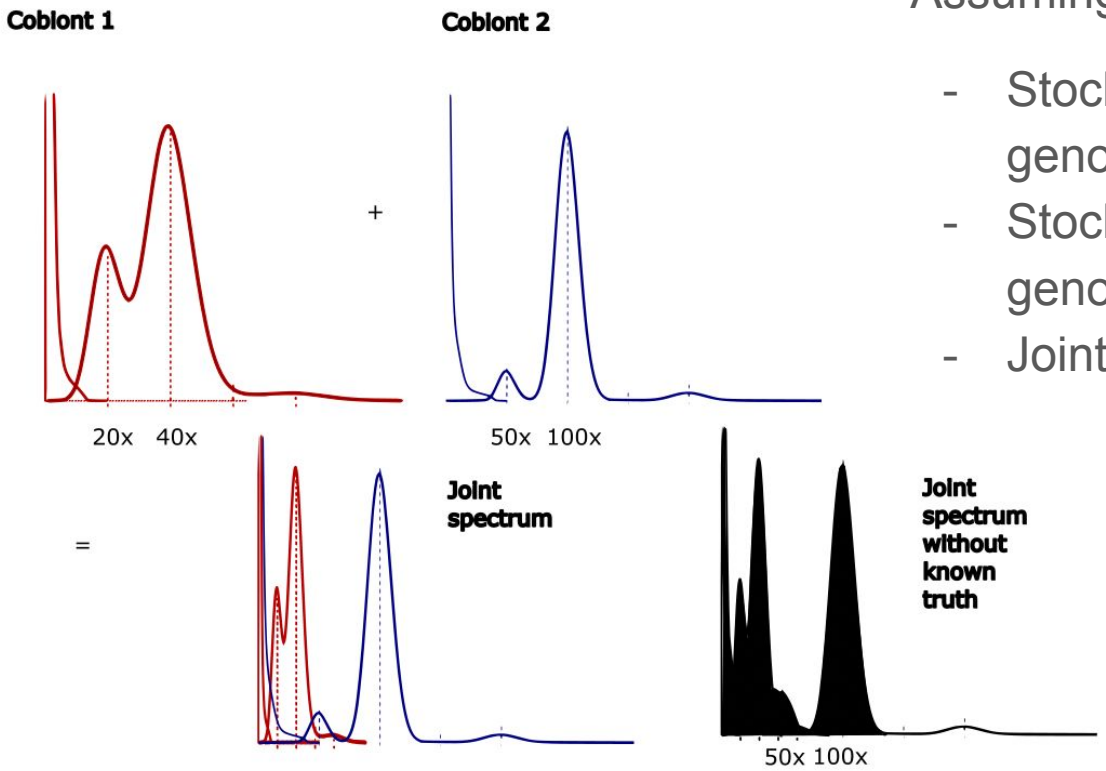
Obligate cobionts



Spectrum of two diploid genomes with very different coverages and genome sizes



Expectations from the co-biont spectra



Assuming very distinct contributors!

- Stochiometry is retained within genomes
- Stochimetry is violated between genomes
- Joint spectrum can be a bit messy

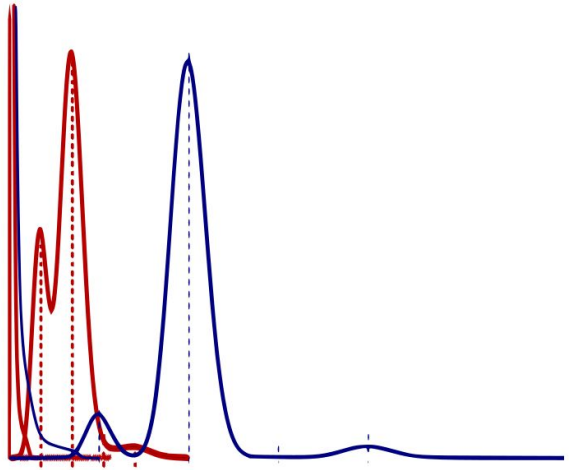
Idea behind cobiont model

Originally

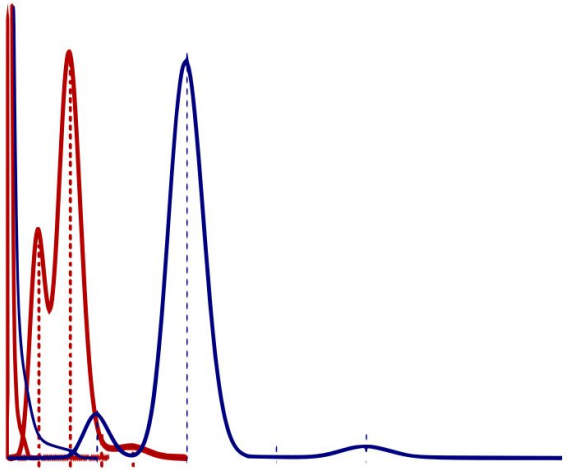
```
Frequency ~ (α NB(  kcov,   kcov / ξ)  +
            β NB(2*kcov, 2*kcov / ξ)) * length
```

The adjustment conceptually

```
Frequency ~ (α NB(  kcov,   kcov / ξ)  +
            β NB(2*kcov, 2*kcov / ξ)) * length
                        +
            (α NB(  kcov,   kcov / ξ)  +
             β NB(2*kcov, 2*kcov / ξ)) * length
```



Separate parameters of the two models



The adjustment

$$\text{Frequency} \sim (\alpha\ NB(\ kcov1,\ kcov1\ /\ \xi)\ +\ \beta\ NB(2*kcov1,\ 2*kcov1\ /\ \xi))\ *\ length1\ +\ (\gamma\ NB(\ kcov2,\ kcov2\ /\ \xi)\ +\ \delta\ NB(2*kcov2,\ 2*kcov2\ /\ \xi))\ *\ length2$$

kcov1, **kcov2** - respective coverages

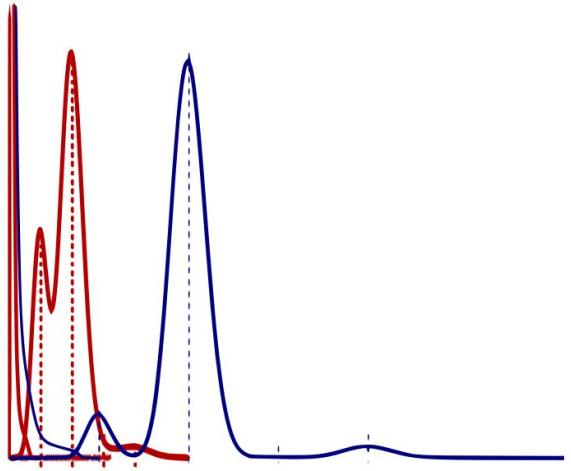
length1, **length2** - respective genome lengths

$\alpha + \beta = 1$

$\gamma + \delta = 1$

ξ - overdispersal; single parameter (scales with coverage)

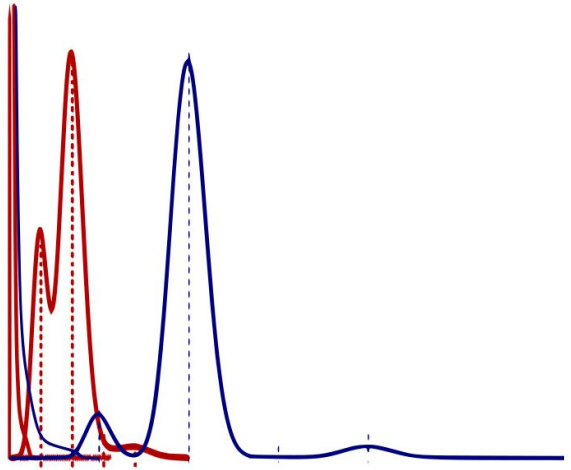
Co-biont model in R



In R (if both haploid):

```
lichen_model <- nlsLM(y ~ (dnbinom(x, mu = kcov, size = kcov/overdispersion) * length1) +
                        (dnbinom(x, mu = kcov2, size = kcov2/overdispersion) * length2),
                      start = list(kcov = kmerEst, kcov2 = kmerEst2,
                      overdispersion = 0.5, length1 = lengthEst, length2 = lengthEst),
                      control = list(minFactor = 1e-12, maxiter = 40))
```

Co-biont model in R



In R (if one is diploid):

```
lichen_model <- nlsLM(y ~ (a       * dnbinom(x, mu = kcov,   size =   kcov/overdispersion) +
                           (1 - a) * dnbinom(x, mu = 2*kcov, size = 2*kcov/overdispersion)) *
                            length1 +
                          dnbinom(x, mu = kcov2, size = kcov2/overdispersion) * length2,
                        start = list(kcov = kmerEst, kcov2 = kmerEst2, a = 0.3,
                        overdispersion = 0.5, length1 = lengthEst, length2 = lengthEst),
                        control = list(minFactor = 1e-12, maxiter = 40))
```

Important bottleneck - very sensible starting values

- Explain singular gradient here

How the non-linear least square method works?

- Minimising residual sequares (similarily to any other linear model you know)
- No exact solution, so you guess some values, calculate gradients for changinge those values and iteratively change them to minimise residual squares
- Gauss Newton method

Estimating the size of heterogametic sex chromosomes



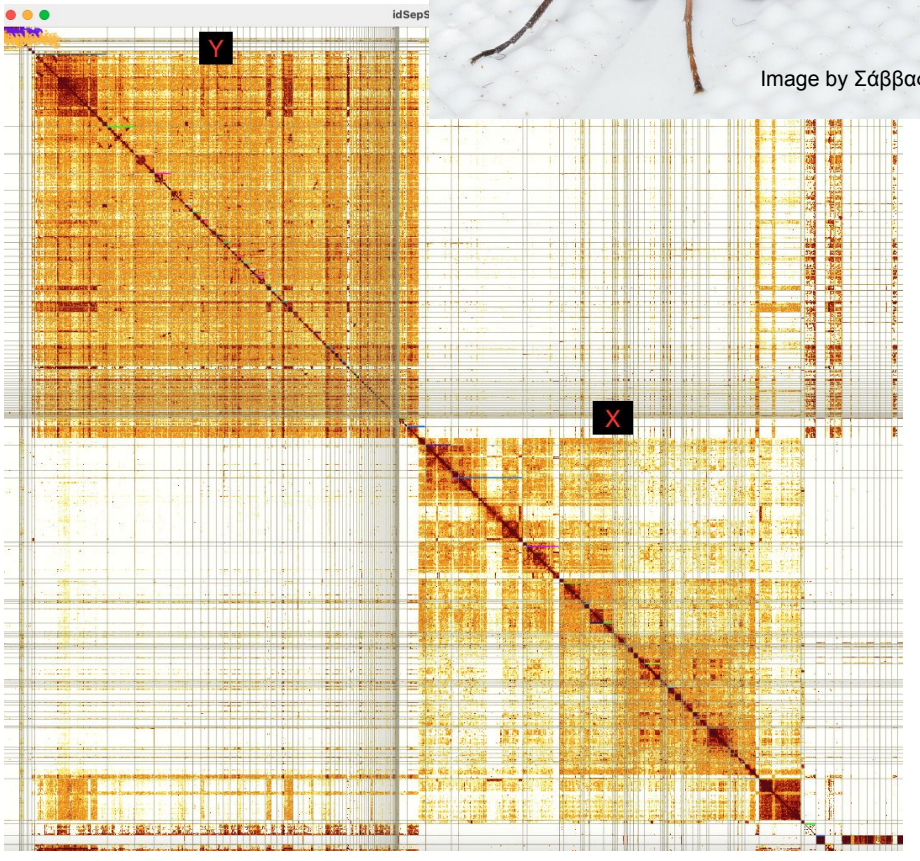
A march fly

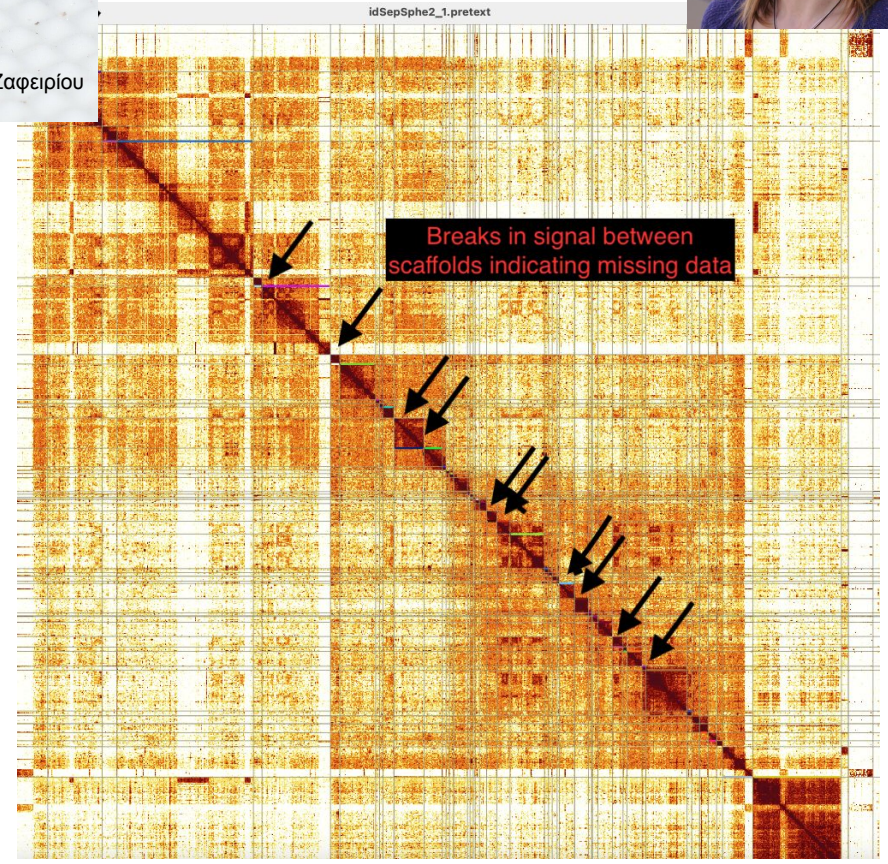


Image by Σάββας Ζαφειρίου

Curated by Joanna Collins







XY sex chromosome model

Workflow for XY species:

1. Assume heterozygosity / heterogametic size (possible to est from homogametic sex)
2. Fit to heterogametic sex k-mer spectra with a fixed parameter & infer the other

```
Frequency   ~ (α NB(  kcov,   kcov / overdispersion)  +
               β NB(2*kcov, 2*kcov / overdispersion)) * (length - XY_len)  +
                 NB(  kcov,   kcov / overdispersion)  * XY_len
```

autosomes

X and Y chromosomes

Bold parameters are fixed constants (fixed parameters estimated in female)

Sex chromosome model - XY 4peaks

```
Frequency   ~ (α NB(  kcov,   kcov / bias)  +
               β NB(2*kcov, 2*kcov / bias)  +
               γ NB(3*kcov, 3*kcov / bias)  +
               δ NB(4*kcov, 4*kcov / bias)    ) * length_diplo +
              (ε  NB(  kcov,   kcov / bias) +
               ζ  NB(2*kcov,   2*kcov / bias)   ) * length_mono
```

autosomes

X and Y chromosomes

If we would like to make it fully “GenomeScope”-like

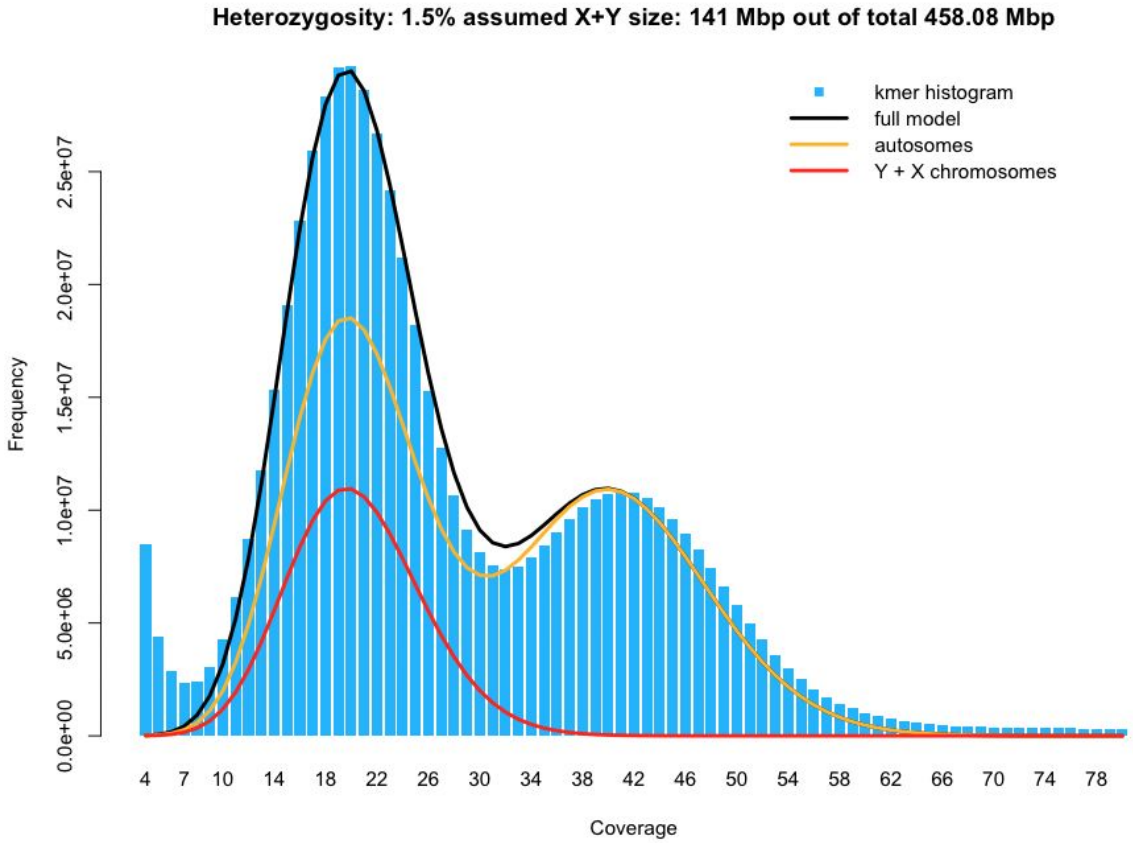
Let’s stick with XY sex chromosome model

```
Frequency   ~ (α NB(  kcov,   kcov / overdispersion)  +
               β NB(2*kcov, 2*kcov / overdispersion)) * (length - XY_len)  +
                 NB(  kcov,   kcov / overdispersion)  * XY_len
```

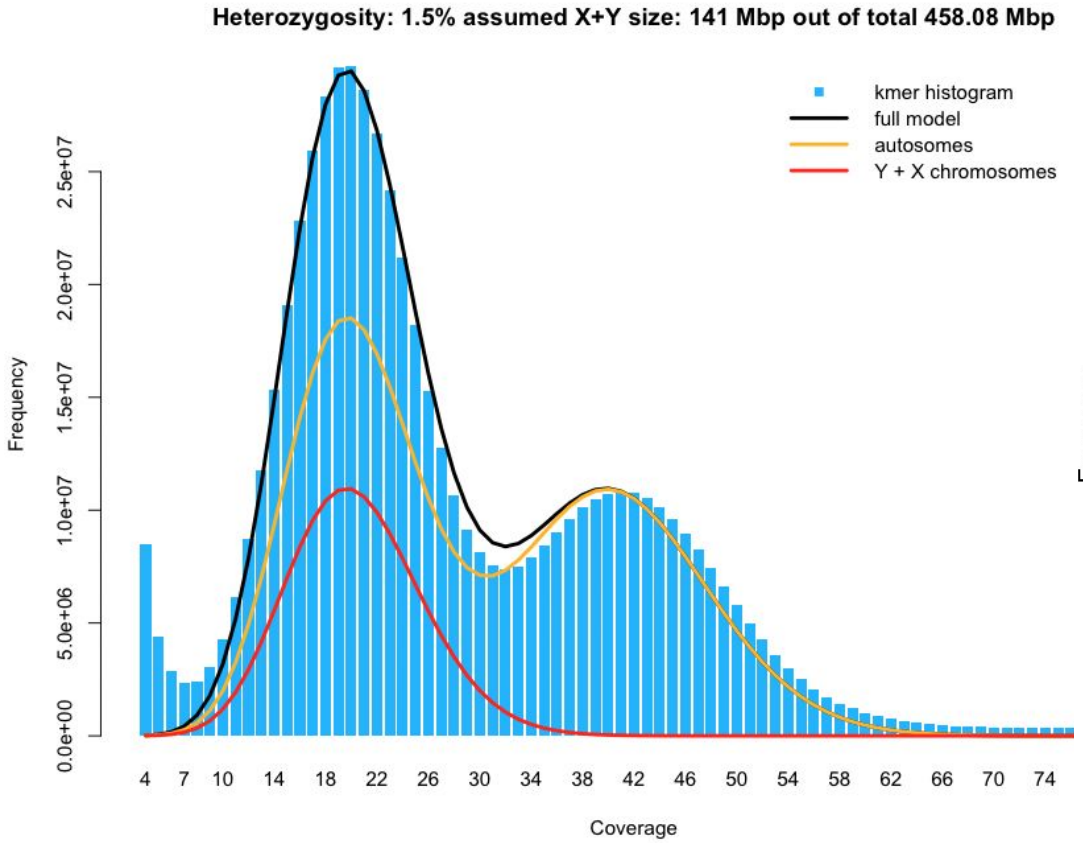
autosomes

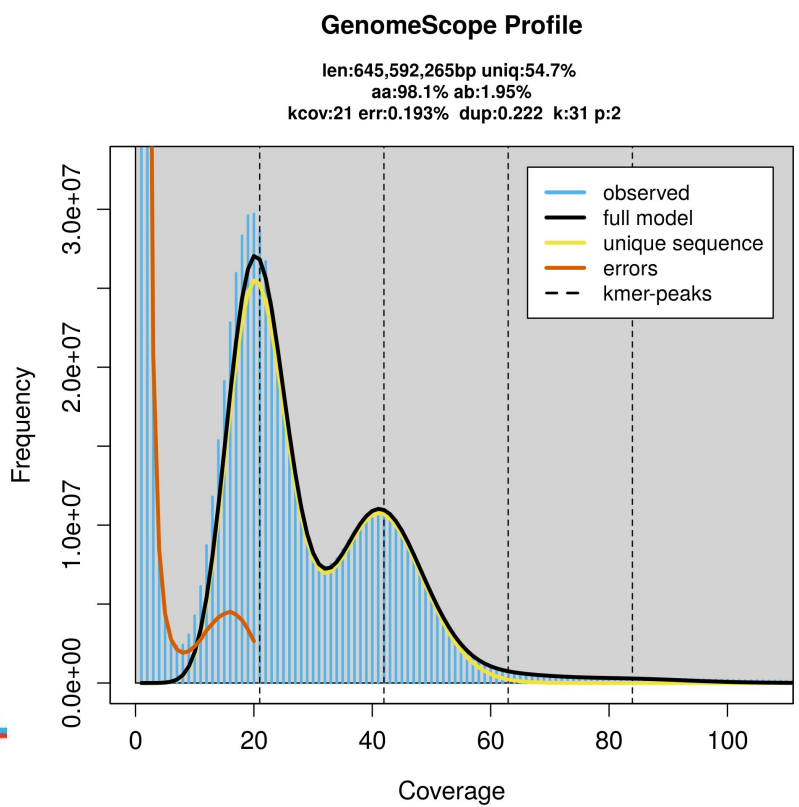
X and Y chromosomes

Estimating spectra composition by fixing heterogametic size

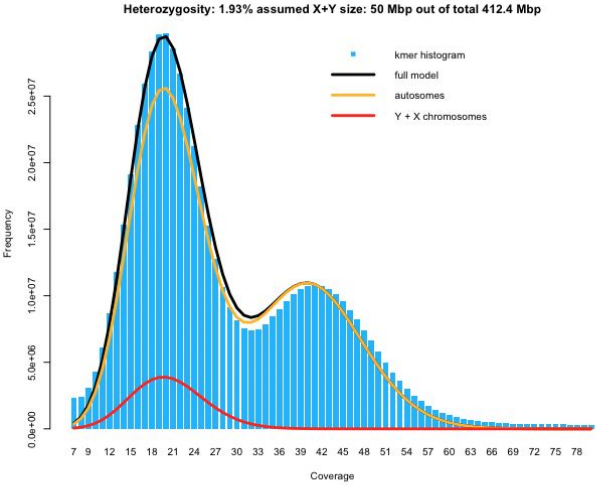


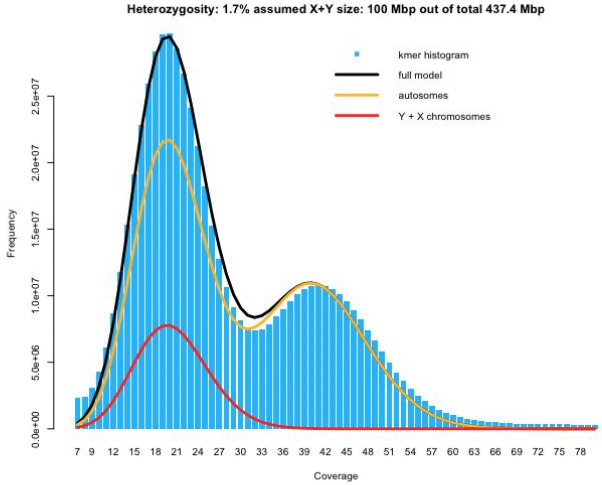
Estimating spectra composition by fixing heterogametic size

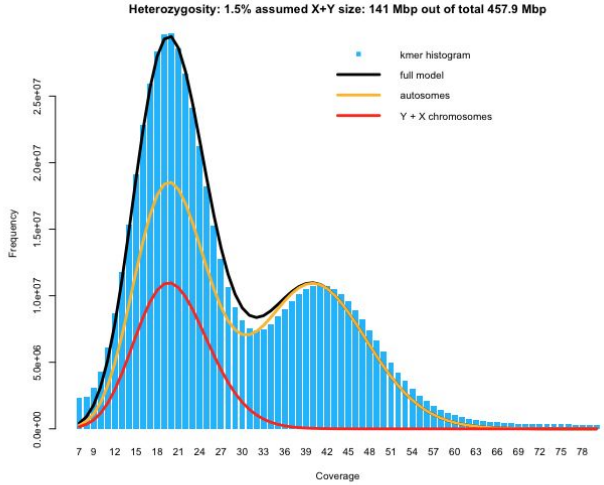


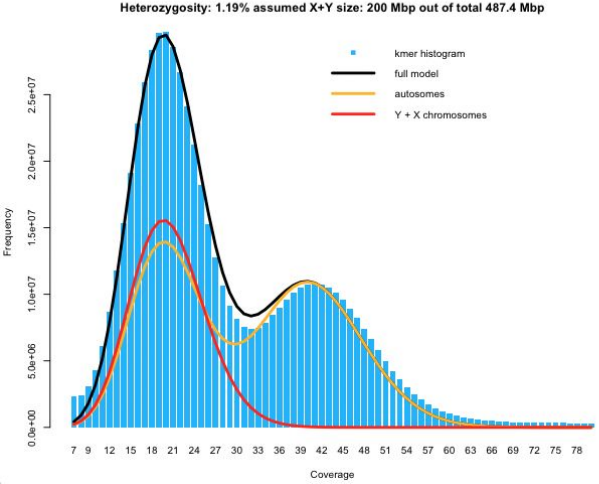


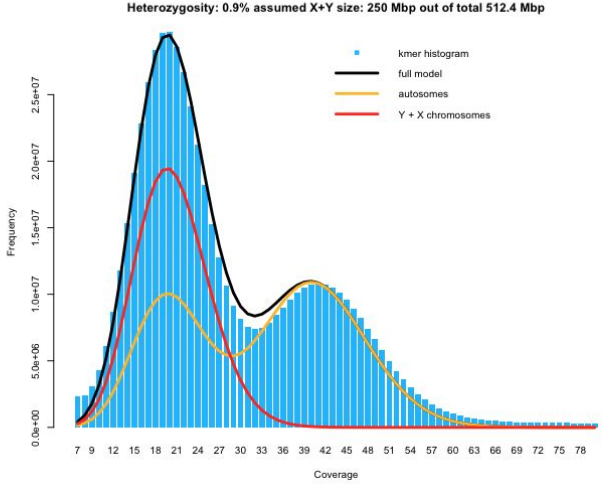
How heterozygous given heterogametic size?

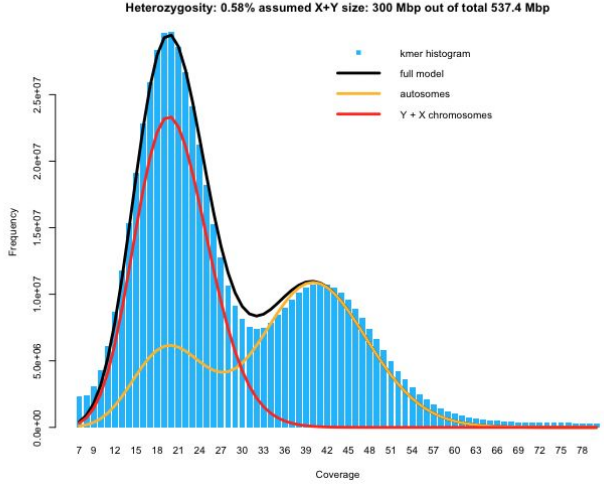






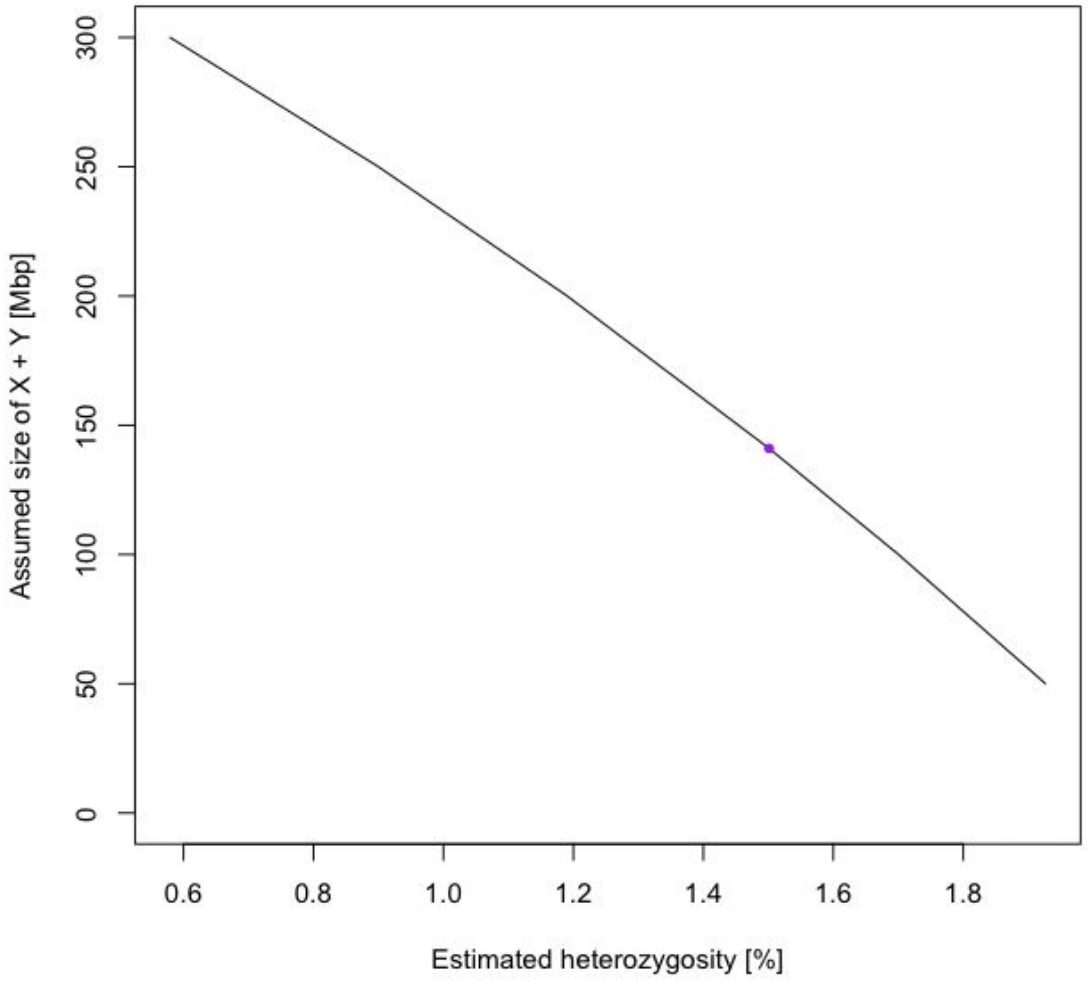




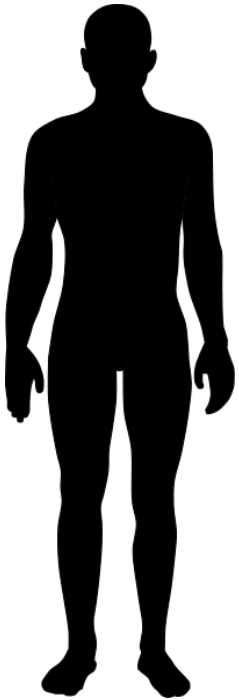


Heterogametic size ~ heterozygosity

- Gives meaningful range of heterogametic sizes / heterozygosities



Testing the same principle on the HG002 human



T2T based

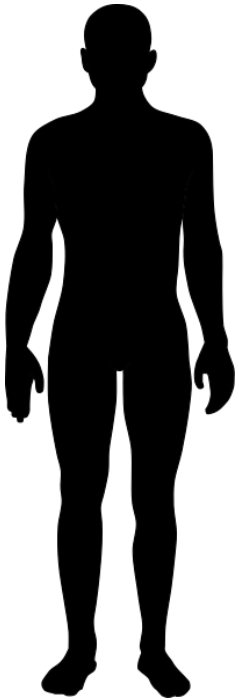
Total
3.11 Gbp

Disomic
2.9 Gbp

Monosomic
0.21 Gbp



Testing the same principle on the HG002 human

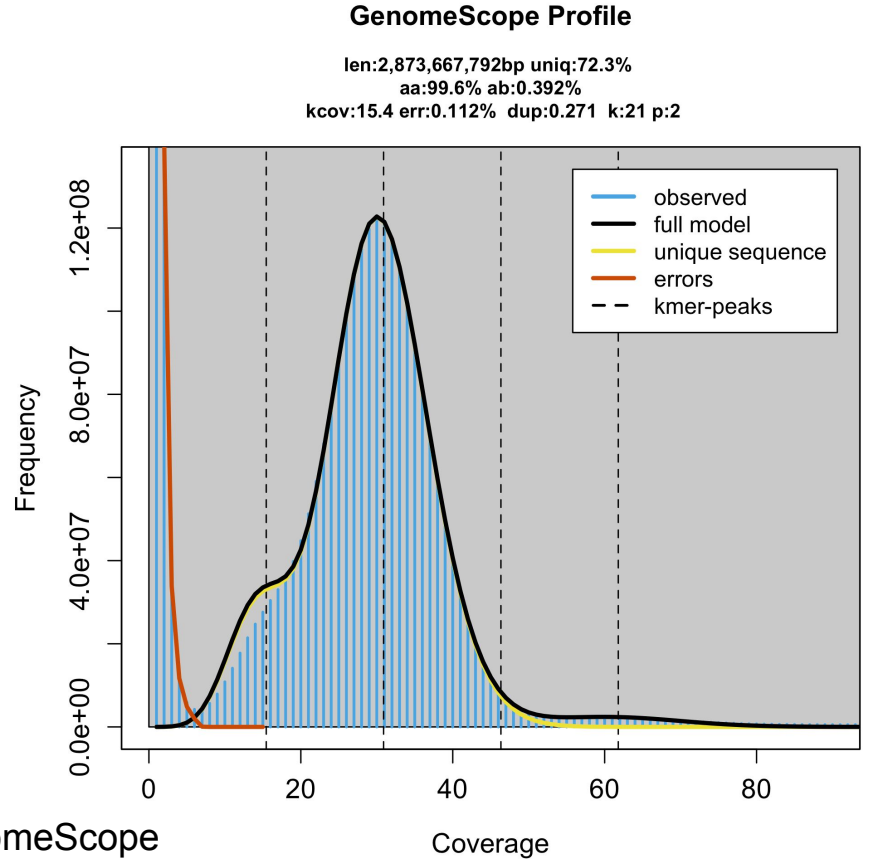


T2T based

Total
3.11 Gbp

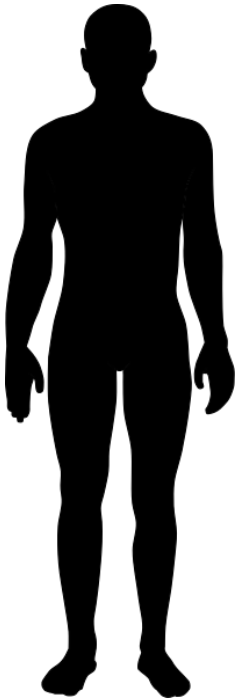
Disomic
2.9 Gbp

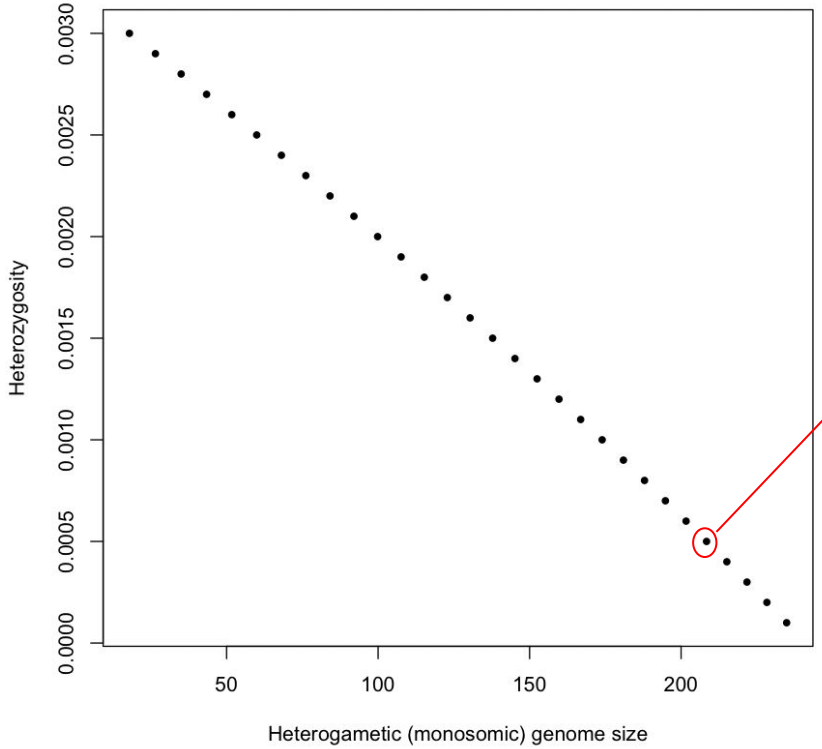
Monosomic
0.21 Gbp



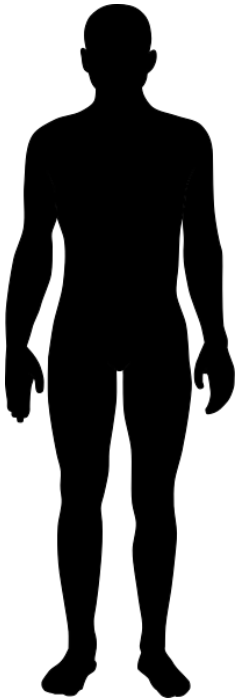
“default” GenomeScope

Given heterozygosity fit a model





Closest to the empirical size



Fitted / Real values

Total

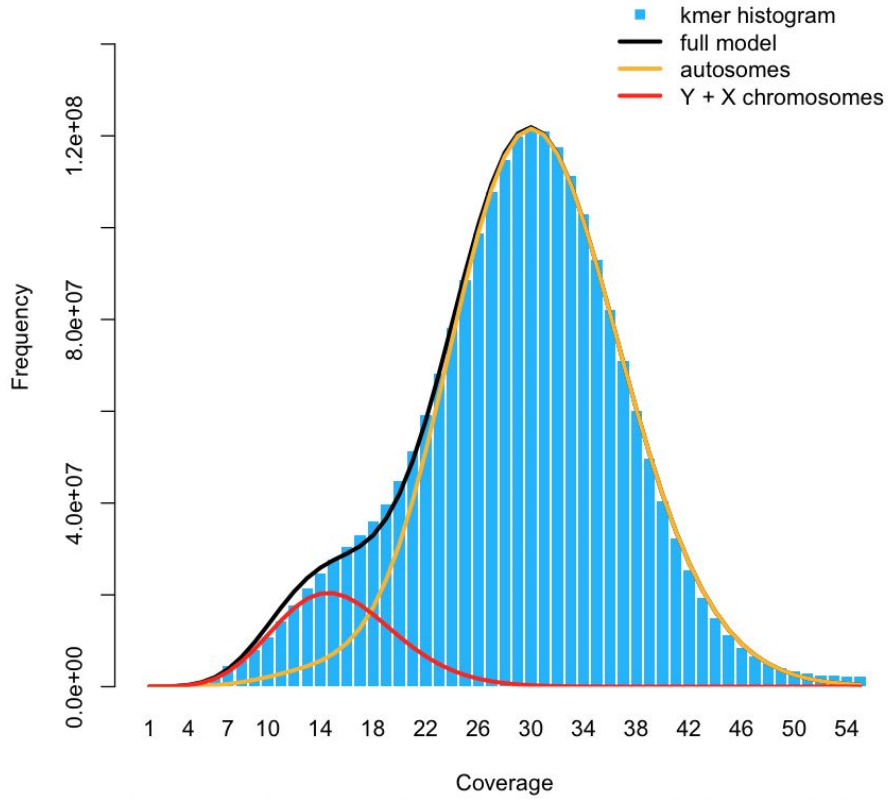
3.03 / 3.11 Gbp

Disomic

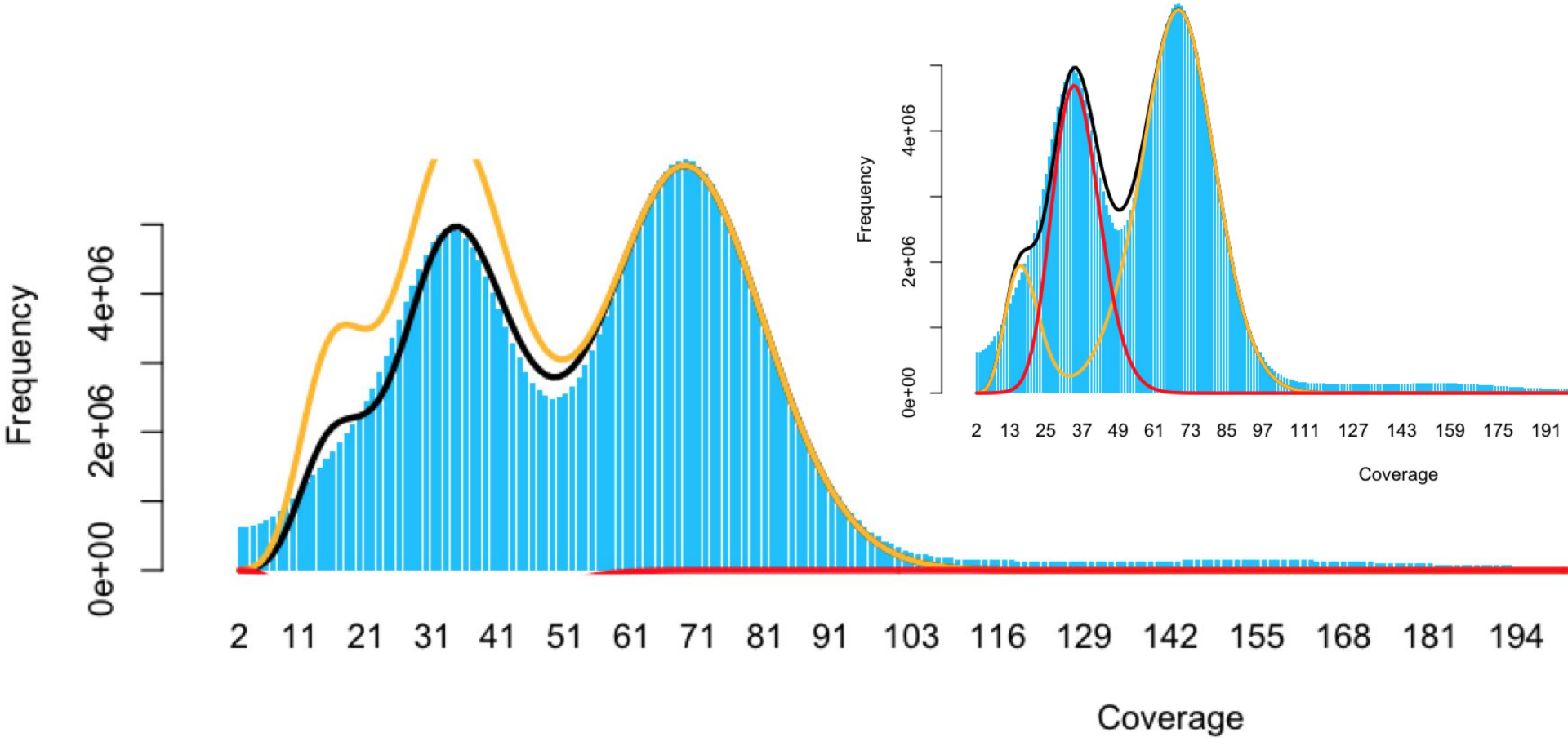
2.75 / 2.9 Gbp

Monosomic

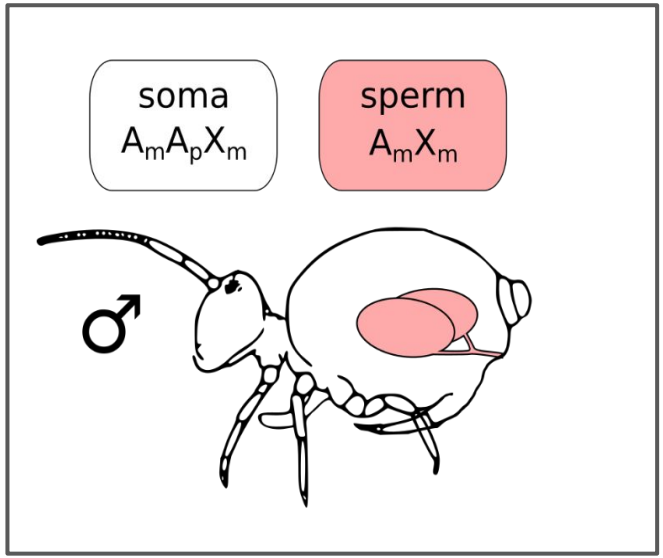
0.288 / 0.21 Gbp



Convergence problems of more complex models

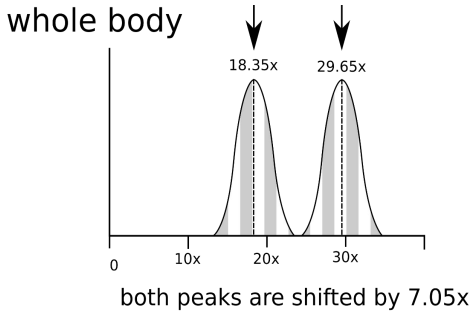


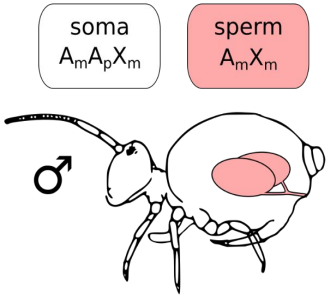
Bonus slides
Mixed karyotypes



Two tissue model

PGE Male





Need to model the 1n and 2n coverage as independent variables

```
Frequency ~ ( α      * NB(kcov,  kcov  / bias)  +
              β      * NB(kcov2, kcov2 / bias)) * length
```

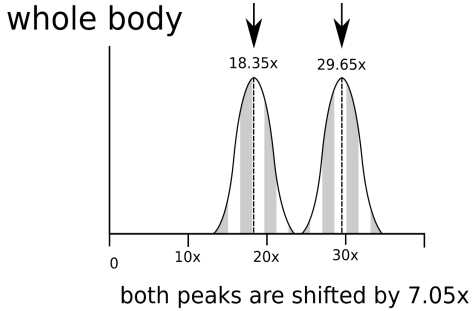
$$f_h = 1 - \frac{(c_A - c_X)}{c_X}$$

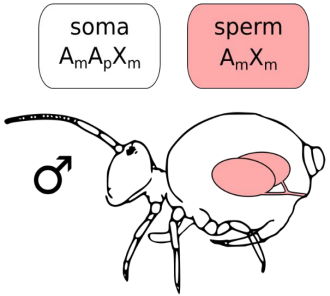
c_A - autosomal coverage peak
(2n peak, kcov2)

c_X - X-chromosome coverage peak
(1n peak, kcov)

Two tissue model

PGE Male





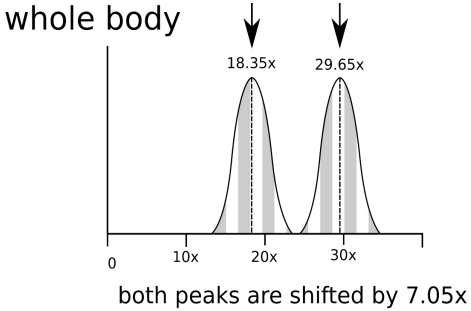
Need to model the 1n and 2n coverage as independent variables

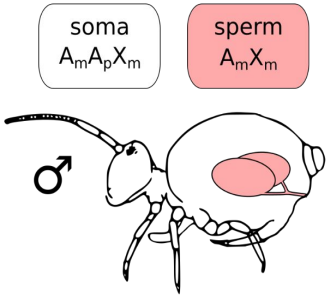
```
Frequency ~ ( α       * NB(kcov,  kcov  / bias)  +
              (1 - α) * NB(kcov2, kcov2 / bias)) * length
```

Modeled as **α** as we don't care about the actual heterozygosity

Two tissue model

PGE Male





Need to model the 1n and 2n coverage as independent variables

```
Frequency ~ ( α       * NB(kcov,  kcov  / bias)  +
              (1 - α) * NB(kcov2, kcov2 / bias)) * length
```

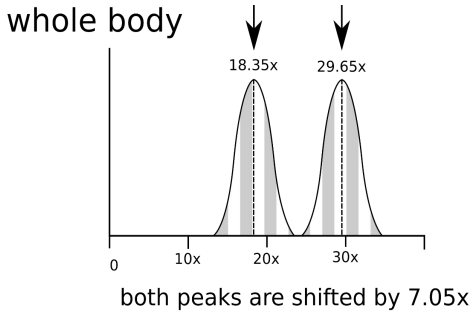
Modeled as **α** as we don't care about the actual heterozygosity

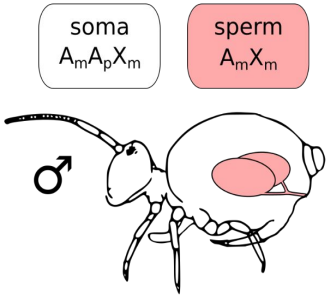
kcov and **kcov2** are the important modeled parameters

bias is still a single parameter in the model :-)

Two tissue model

PGE Male





Need to model the 1n and 2n coverage as independent variables

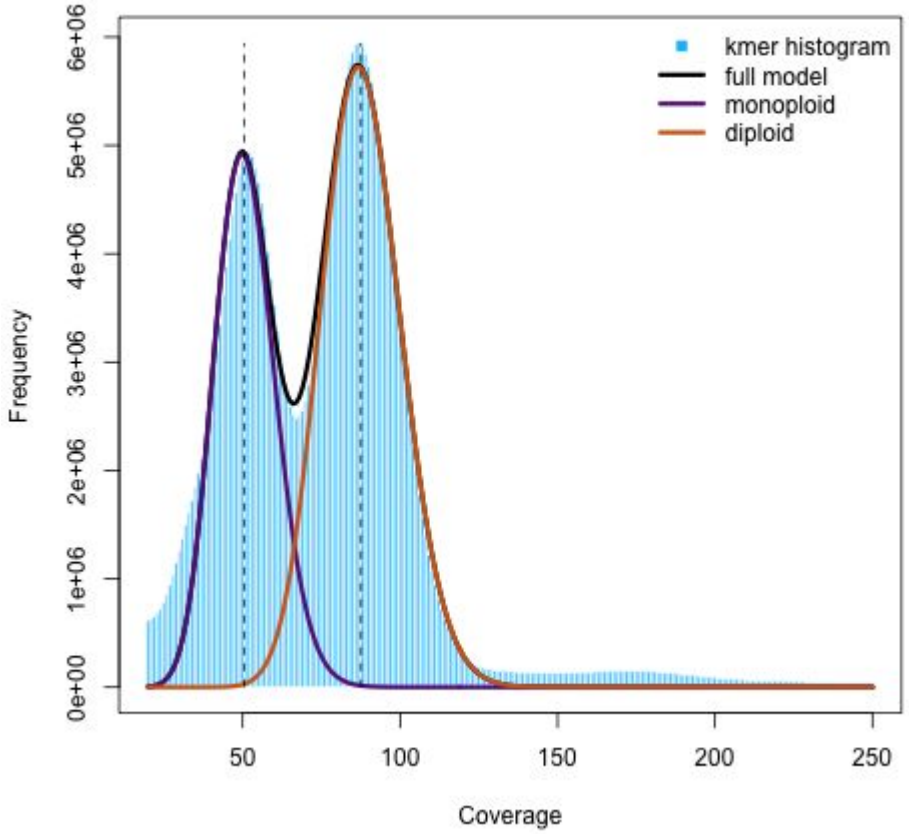
```
Frequency ~ ( α       * NB(kcov,  kcov / bias)  +

              (1 - α) * NB(kcov2, kcov2 / bias)) * length
```

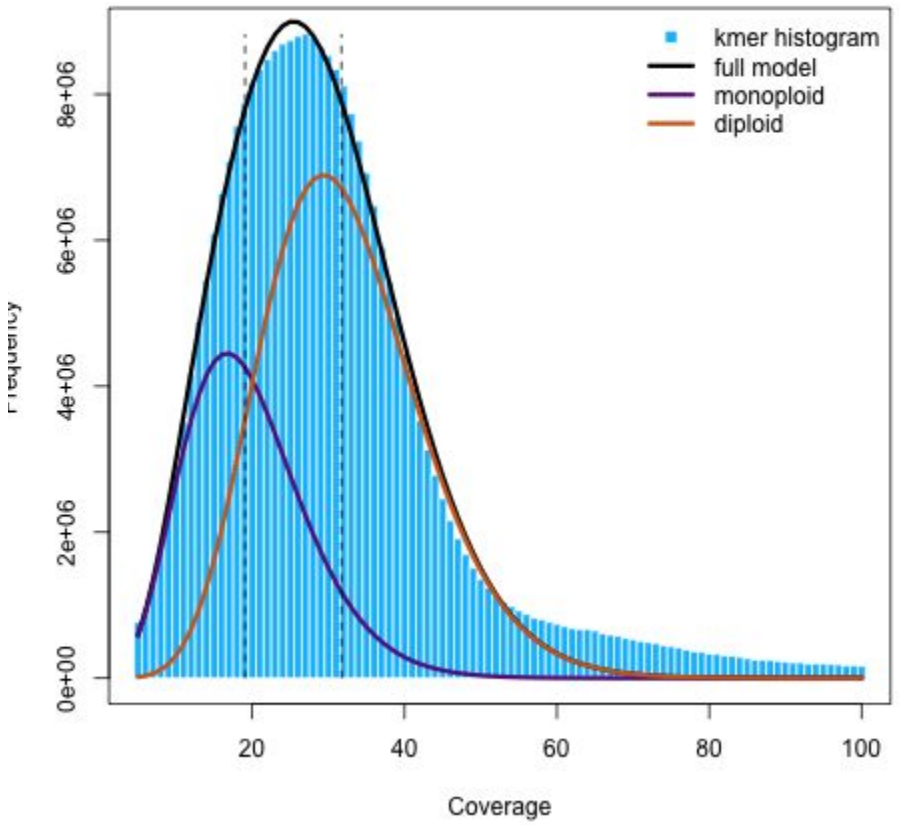
```
kmerEst <- 50
PGE_model <- nlsLM(y ~ ((het         * dnbinom(x, size = kmercov   / bias, mu = kmercov)) +
                        ((1 - het)   * dnbinom(x, size = kmercov2  / bias, mu = kmercov2))) * length,
                   start = list(kmercov = kmerEst, kmercov2 = (2*kmerEst), bias = 0.5, length = 280e6, het = 0.6),
                   control = list(minFactor=1e-12, maxiter=40))
```

Simple PGE model fit the k-mer spectra really well

Afus1

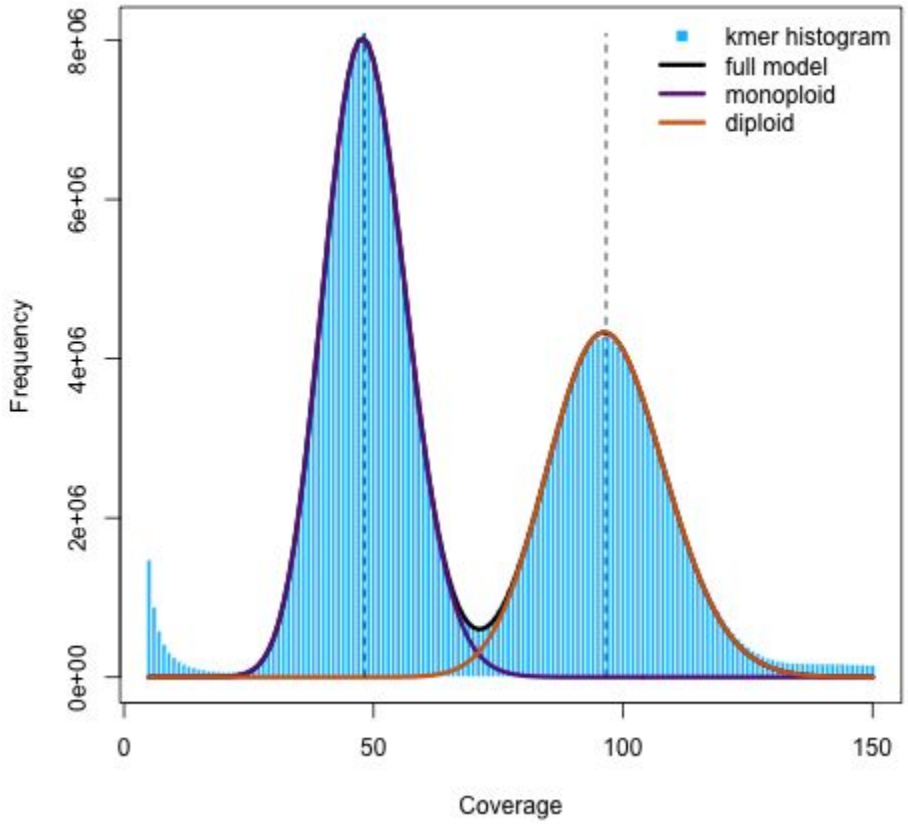


BH3-2



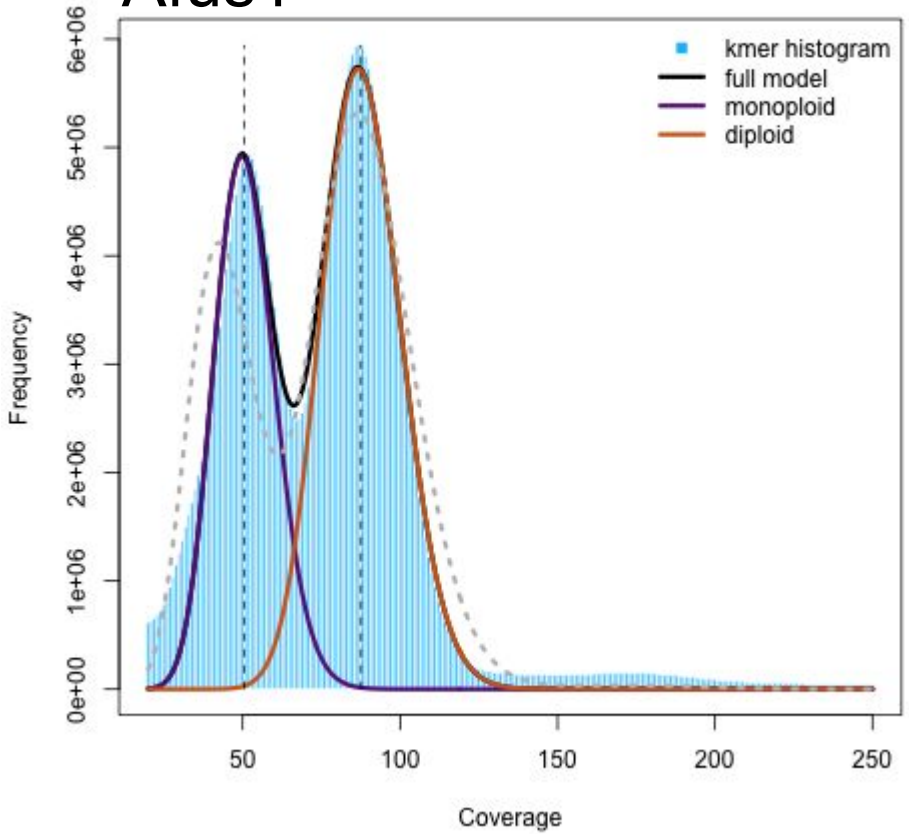
Simple PGE model fit the k-mer spectra of non-PGE species really well too

Ocin2

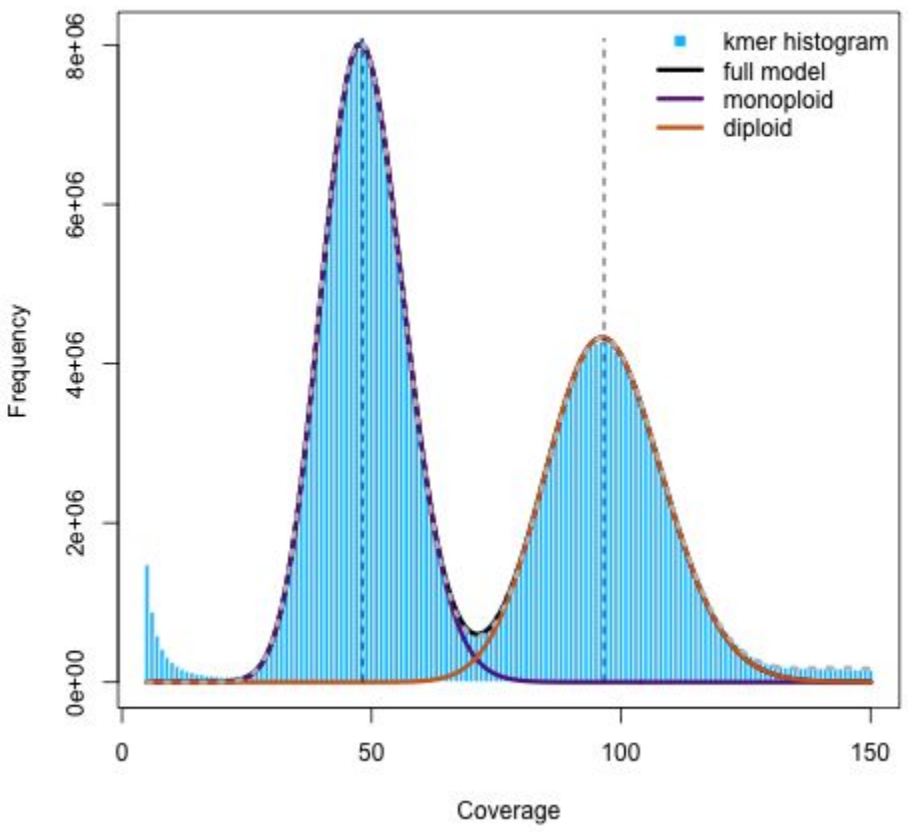


GenomeScope (gray dashed line) vs Simple PGE fits

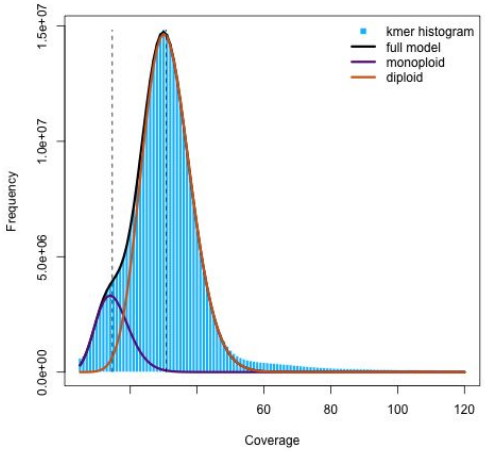
Afus1

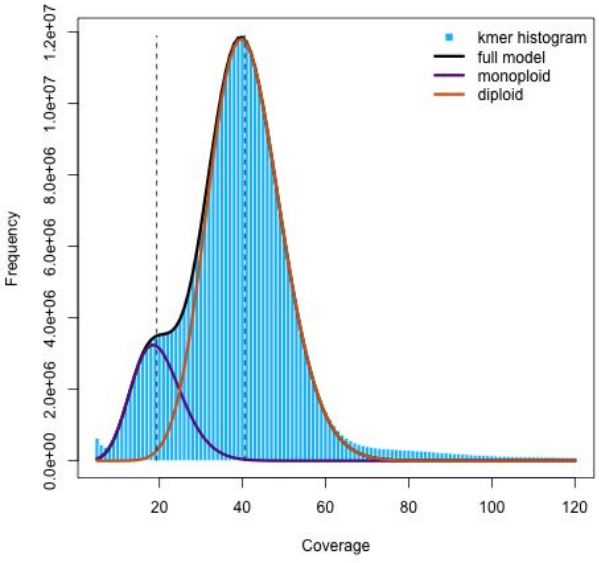


Ocin2



Females


kmer histogram
full model
monoploid
diploid
Frequency
0.0e+00
5.0e+06
1.0e+07
1.5e+07
60
80
100
120
Coverage


kmer histogram
full model
monoploid
diploid
Frequency
0.0e+00
2.0e+06
4.0e+06
6.0e+06
8.0e+06
1.0e+07
1.2e+07
20
40
60
80
100
120
Coverage

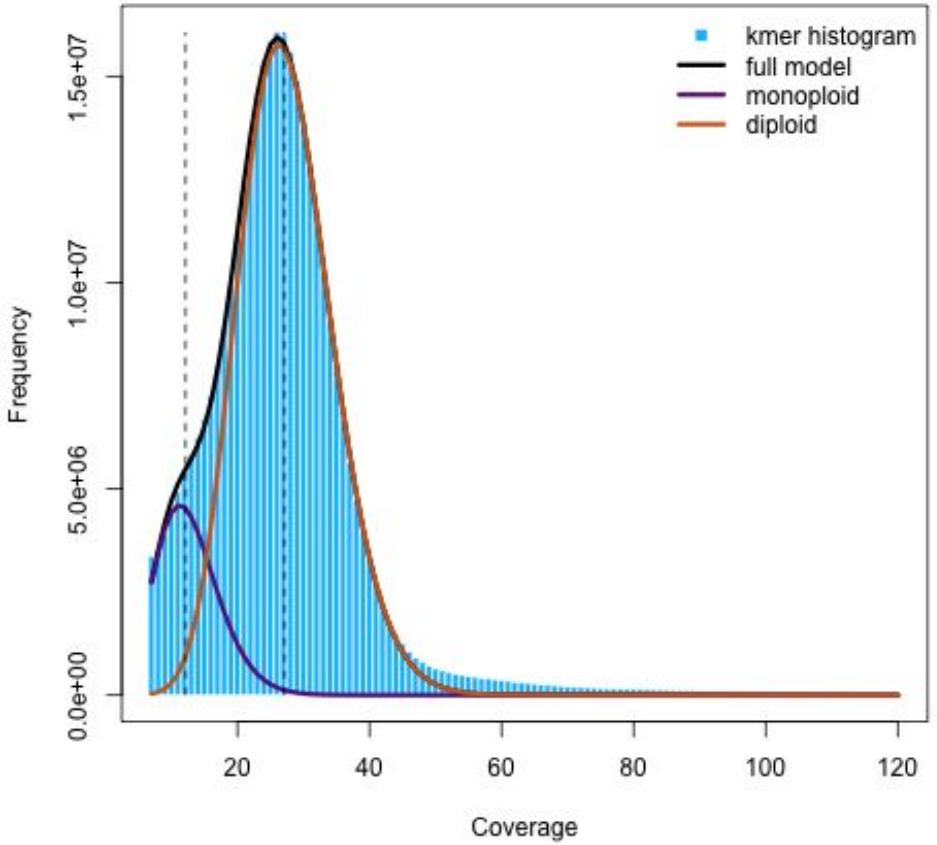
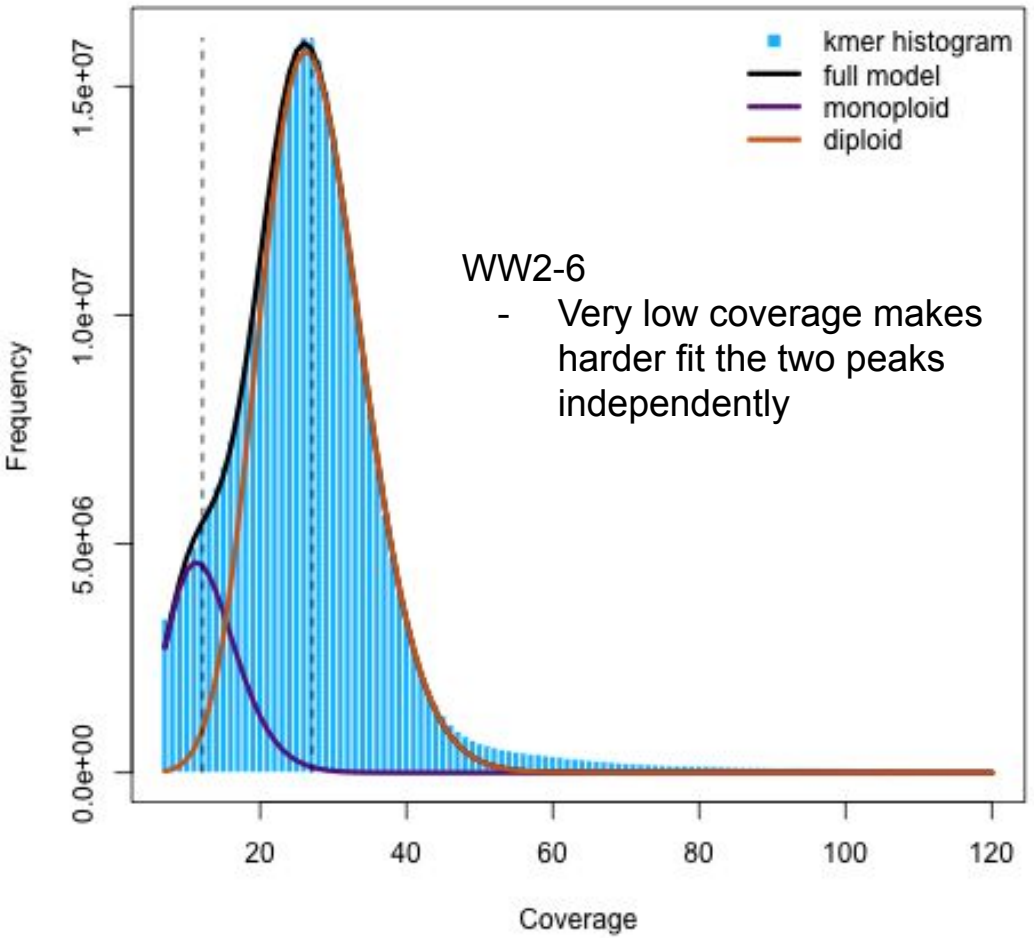

kmer histogram
full model
monoploid
diploid
Frequency
0.0e+00
5.0e+06
1.0e+07
1.5e+07
20
40
60
80
100
120
Coverage

Table of 1n and 2n est

ind	cov_1n	cov_2n
Afus1	51.05	88.05
BH3-2	19.55	32.21
Ocin2	48.68	97.09
WW5-3	15.11	31.23
WW5-5	19.85	41.05
WW2-6	12.61	27.59

Table of 1n and 2n est

ind	cov_1n	cov_2n	Estimated sperm
Afus1	51.05	88.05	0.275
BH3-2	19.55	32.21	0.352
Ocin2	48.68	97.09	0.005
WW5-3	15.11	31.23	-0.066
WW5-5	19.85	41.05	-0.068
WW2-6	12.61	27.59	-0.232



WW2-6

- Very low coverage makes harder fit the two peaks independently

Estimated sperm
0.275
0.352
0.005
-0.066
-0.068
-0.232

Explicit PGE model

```
nls(y ~ ((1-(1-r)^k) * dnbinom(x, size = kmercov_p / bias, mu = kmercov_p) +
          (1-(1-r)^k) * dnbinom(x, size = kmercov_m / bias, mu = kmercov_m) +
          ((1-r)^k)   * dnbinom(x, size = (kmercov_p + kmercov_m) / bias, mu = kmercov_p + kmercov_m)) *
           length * fraction_diploid +
          dnbinom(x, size = kmercov_m / bias, mu = kmercov_m)                    *
                                                 length * (1 - fraction_diploid),
    start = list(kmercov_p=estKcov, kmercov_m=estKcov, bias=bEst, length=estLen, fraction_diploid =
estF),
    control = list(minFactor=1e-12, maxiter=max_iterations))
```

heterozygous

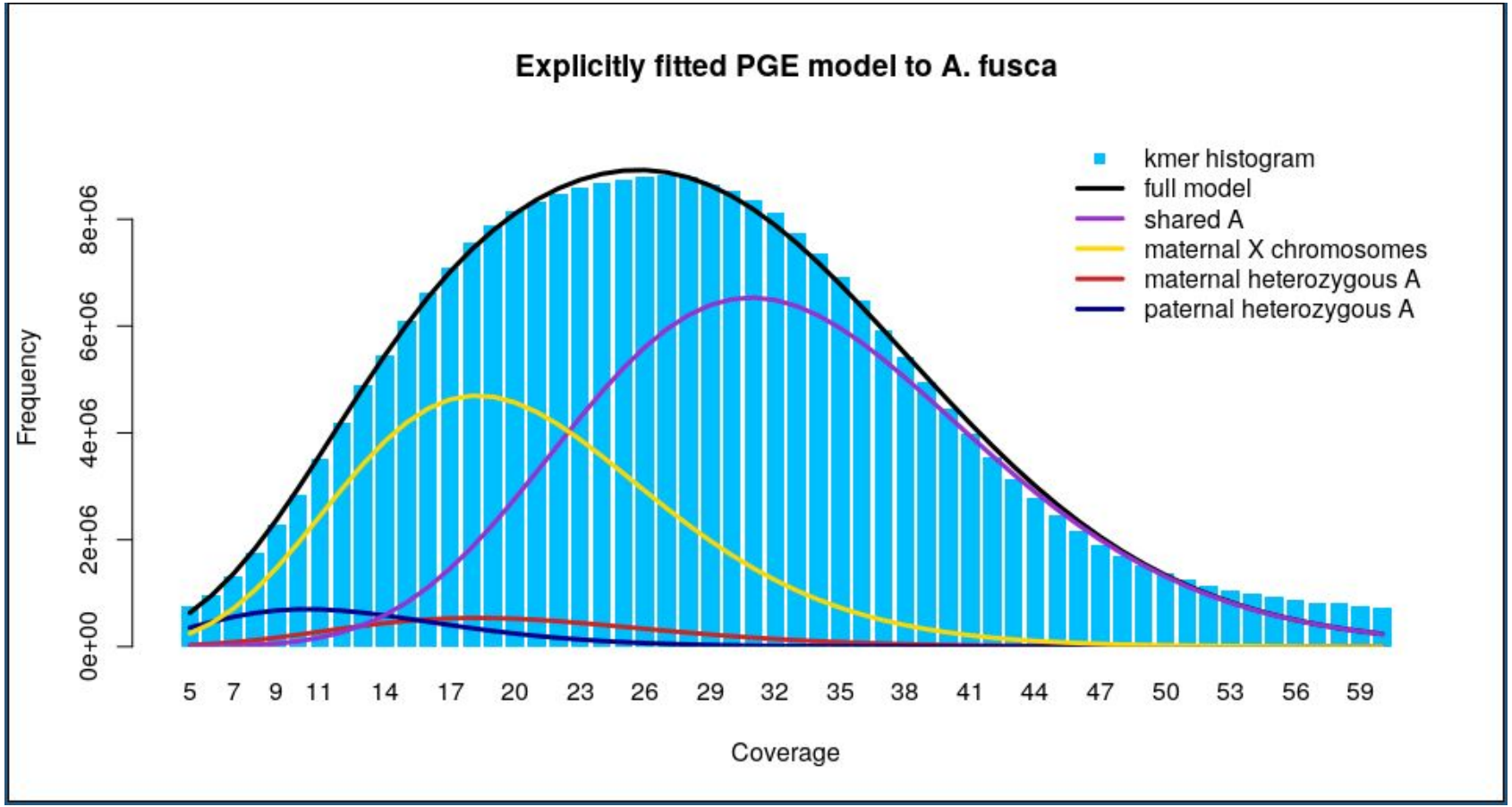
heterozygous

homozygous

Autosomal

X chromosome(s)

Explicit PGE model fit to BH3-2



Summary

- K-mer spectra is an extremely efficient and powerful way to look at genomic data
- We can construct alternative genome models. Some pointers:
  - keep the parametric space sane
  - co-estimate parameters whenever we can, fix parameters when not
  - model fit is extremely dependent on initial values

- We are developing **GenomeTelescope** to facilitate the alternative genome models
  - https://github.com/KamilSJaron/GenomeTelescope
  - Two Tissue Model could potentially tell us about GRCs already, but likely we will need a better model for that